Analysing the Behaviour of Different Types of Beams with Finite Element Models

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By

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This thesis was prepared for the School of Civil Engineering at the University of Sydney, Australia, and describes the necessity for engineers to use the correct model for the design scenario at hand through comparison of the output of different finite element models. The opinions, conclusions and recommendations presented herein are those of the author and do not necessarily reflect those of the University of Sydney or any of the sponsoring parties to this project.

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Summary

Finite element analysis is a popular tool used by engineers to aid in the design of structures. However, the underlying assumptions of these models may cause them to underestimate beam stresses and displacements if they are used in a situation where these assumptions are not valid. This thesis presents the importance of using a finite element model that is suitable for the design scenario at hand. Models were developed in Matlab based on linear Euler-Bernoulli beam theory (EBBT) and Timoshenko beam theory (TBT) as well as nonlinear Euler-Bernoulli beam theory.

A beam was analysed using the linear models to show the differences in the output of each model for differing length to depth ratios. It was found that EBBT significantly underestimated the displacements experienced by a deep beam, however, this error reduced as the length to depth ratio was increased. Additionally, it was found that for a high length to depth ratio, EBBT and TBT provided very similar results, but EBBT could be analysed using a much coarser mesh and was therefore more computationally efficient. It was also deduced that using nonlinear analysis provided a more cost-effective solution as it allows the full capacity of a beam to be realised, therefore allowing smaller members to be used. Finally, the dangers of using a linear model once the material has yielded was shown as the linear EBBT significantly underrepresented the beam displacements once the yield strain was surpassed.

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Chapter 1: Introduction

The design of structures requires an engineer to determine how members are sized and arranged to ensure that strength and serviceability requirements are met. To do this, detailed calculations must be completed to determine the structural response of the members due to the design loads. One significant structural element that must be designed is a beam. Beam elements behave differently to an applied load depending on their dimensions, including their length to depth ratio and thickness.

Majority of beams have a length much greater than their depth. These types of beams are dominated by flexural deformations and therefore have negligible cross-sectional deformations and shear stresses. Such beams are simple to model as assumptions can be made that greatly reduce the complexity of the required analysis.

Open spaces are becoming increasingly prevalent in the building industry. Having open spaces has been found to enhance collaboration in workplaces and allows for greater flexibility in the use of a building. Additionally, there are structures such as shopping centres where having a significant number of columns would be undesirable as large open areas are required. However, open spaces require long beams, meaning that they must be deep to achieve an adequate stiffness for strength and serviceability requirements. Another common use of deep beams is for bridges, which are required to carry large loads over a great distance. Deep beams are dominated by shear deformations which must be accounted for to accurately describe their behaviour. As a result, analysis of these beams will require a more complicated analysis than flexural dominated beams.

Some structural members are composed of extremely thin elements, known as thin-walled members (TWM) which have a low ratio of thickness to width. One popular example of a TWM is cold formed steel, which is widely used in the construction industry for roof purlins and girts. During manufacturing, the steel is passed through a series of rollers at a normal temperature until it is typically between 0.4mm and 6.4mm thick. This process makes the steel light with high strength but also makes it highly susceptible to buckling. When a member buckles, it experiences as sudden sideways deflection. Buckling is a dangerous failure mode as it will often occur without any warning and before it yields, which is why an accurate buckling analysis is crucial when designing TWMs. This further complicates the analysis of these members as while the previous types of beams only require analysis of the member deformations, the cross-sectional deformations must also be analysed in this case.

The discussion thus far has been relevant to beams that are loaded within the linear elastic range of the material that the beam is composed of. However, when a beam experiences a strain greater than its yield strain, a nonlinear material model must be used to analyse its behaviour. Plastic analysis is useful because the full capacity of a structure can be realised. This is because once yield is achieved, the loads are redistributed to the areas of the cross section that usually don't experience a significant amount of loading. This analysis allows the load to be determined that results in failure, which is much greater than the load that causes yield. As a result, it is possible to select a smaller member size for design which allows for a more cost-efficient solution to be developed.

Finite element analysis (FEA) software is a useful tool for the analysis of structures as it allows complicated analysis to be completed in a short amount of time. This allows a designer to investigate the behaviour of complex structures that are impossible to analyse by hand. FEA requires the structure to be divided into elements of a chosen size, orientation, and type. These elements are connected to the adjacent elements by nodes, allowing for continuity throughout the member. Once the contribution of each element is calculated, these contributions can be synthesised to determine the behaviour of the entire member.

Each finite element model is developed based on underlying assumptions that vary depending on the type of member that is intended to be analysed. These models are only valid for structures for which these assumptions are met. As a result, a model created to analyse a flexural dominated beam would not be valid for a TWM. Conversely, because a model created to accurately analyse a TWM would have much fewer simplifying assumptions, it is likely that it would be able to accurately model a shallow beam. However, this would not be efficient as it would be much more computationally intensive and would require more information to be input into the model.

1.1 Research Problem and Proposed Research

Engineers often utilise finite element analysis software when designing a structure to determine whether the given structure can resist the design loads applied to it. As a result, it is crucial that the results output from the model are accurate so that a safe design is achieved. However, it is unlikely that engineers are aware of the assumptions that have been made for these models. There is a significant danger in using finite element software without an understanding of its underlying assumptions and the appropriate scenarios it can be used for. This is because the inappropriate use of a model can lead to significant discrepancies from the real behaviour of a structure.

This thesis aims to highlight the issues related to the use of an inappropriate model by comparing the output of various finite element beam models. One of these models will be based on the Euler-Bernoulli beam theory, which is a simple beam theory capable of analysing beams that are dominated by flexural deformations. Another will be based on the Timoshenko beam theory, which is an extension of the Euler-Bernoulli beam theory that is capable of accounting for shear deformations. The degree to which shear deformations influence the results output by the model will be explored to determine the importance of using the correct beam model. Additionally, the computational efficiency of these models will be compared to see if using a more complicated beam model for a simple scenario would have negative implications for the analysis.

The Euler-Bernoulli beam model will then be extended to account for nonlinear effects using an iterative process called the Newton Raphson method. The degree to which nonlinearity influences the response of the beam will be investigated. Additionally, the increase in the capacity of the beam achieved by completing this analysis will be determined.

Chapter 2: Literature Review:

This chapter will provide a record of the development of the finite element method and beam theories over time. It will introduce the Euler-Bernoulli and Timoshenko beam theories, which are two of the most widely used theories for describing the behaviour of a beam and are the theories that will be utilised in this thesis. Their benefits and shortcomings will be explored to provide a basis of understanding that will assist in the investigation of these theories.

These beam theories have existed for centuries, and their continual popularity is due to their simplicity and ease of use. Therefore, developments made to these theories are limited. As a result, despite not being explored further in this thesis, a more complicated beam theory known as Generalised Beam Theory will be reviewed to show the recent progression in methods used to analyse more complicated beams.

2.1: Finite Element Method

The finite element method (FEM) was first introduced in 1941 by Courant, who worked in the field of applied mathematics. Courant used triangles to discretise the cross section of a hollow shaft. The nodal values of stress were determined and then a chosen stress function was interpolated over each triangle to obtain the stress along the shaft [1]. However, as computers were not widely available during this period, his work did not gather any attention from the community.

Towards the end of World War II, aircraft technology was advancing, and fighter planes were beginning to be designed with swept back wings. This meant that the force method, which was the method of analysis used at the time, was highly inconvenient because the orientation of the wings made calculations impractical [2]. This resulted in poor design and aircraft failures, creating the necessity for a better method of analysis. Although not known as the finite element method at the time, FEM began to be used by aeronautical engineers to model the aircraft wings by discretising them into triangles with three nodes. However, due to the policies of the companies they worked for, little of their work was publicised in papers [2]. Initially, proving the advantages of FEM was difficult due to the limitations of technology at the time. The existing computers were expensive and could only run a limited number of calculations, meaning that refined meshes were not possible. This limited the accuracy of the approximations that were possible using FEM. The rapid development of computers over the decades allowed finite element analysis to be more practical because an increasing number of calculations could be computed, enabling finer meshes to be used thus resulting in more accurate analysis.

Throughout the late 1950s and the 1960s, Argyris, Clough, Martin and Zienkiewicz extended the application of FEM to other fields of engineering [3]. Finite element softwares began to be produced in the late 1960s and early 1970s. For example, in 1971, NASA released a generic structural analysis software called NASTRAN to the public. It continued to be developed over the years and is still used today for a variety of applications including railroad tracks, aircraft, and buildings. Due to the development of finite element software, FEM became much more widely used for design and has grown in popularity exponentially over the decades that followed.

Today, FEM has a wide variety of possible applications within engineering, such as modelling the temperature distribution within solids and fluids, analysis of structures and static analysis of aircraft structures. The accuracy of the solution obtained from finite element analysis depends on the complexity of the underlying model as well as the validity of the assumptions made for the analysis.

The theory chosen should be selected through careful consideration of the application and whether the assumptions underlying the theory are satisfied. This thesis will compare the output of finite element models created using different beam theories.

2.2: Euler-Bernoulli Beam Theory

Euler-Bernoulli beam theory (EBBT), also known as classical beam theory, was proposed in the 1750s by Leonhard Euler and Daniel Bernoulli. During this period, the design of structures was largely based on precedent rather than mathematical calculation, thus, their work was not widely used at the time. However, in the late 19^{th} century, the Euler Bernoulli beam theory was used in the design of the Eiffel Tower, allowing it to gain traction in the community and become a widely used tool for the analysis of beams even today, hundreds of years after its development.

The enduring popularity of EBBT is owed to its simplicity, which is a result of the two major assumptions that form the basis of the beam theory. Firstly, it is assumed that beam deformations are small. The second major assumption is that plane sections remain plane and perpendicular to the member axis after deformation. Consequently, shear strains and distortions are neglected, meaning that out of plane deformations are not considered.

Although these assumptions reduce the accuracy of the results obtained when using this beam theory, the results are generally 'good enough' for design if the assumptions are sufficiently valid for the beam under consideration.

In general, EBBT is most successfully applied to beams that long and slender, with a length to depth ratio greater than 10. This is because long beams are dominated by flexural deformations rather than shear deformations [4]. If a deep beam was to be analysed using EBBT, the deflections of the beam would be underestimated, thus a true understanding of its behaviour would not be achieved.

Recent developments have been made to improve Euler-Bernoulli beam theory. One such example is the creation of a new formulation of EBBT that does not neglect shear deformations. Rather than assuming that the beam has isotropic behaviour, the material response is assumed to be anisotropic elastic [5]. The formulation presented was found to obtain valid results, however, takes longer to converge as the governing differential equations are of a higher order than regular EBBT [5].

2.3: Timoshenko Beam Theory

In 1922, an extension to EBBT was proposed by Stephen Timoshenko and Paul Ehrenfest which is known as Timoshenko beam theory (TBT). This beam theory accounts for shear deformations by relaxing the assumptions of EBBT such that plane sections remain plane after deformations, but do not necessarily remain orthogonal to the member axis.

Timoshenko beam theory is most applicable to beams that have a length to depth ratio less than 10 as these members are substantially influenced by shear deformations [4]. TBT is the simplest beam theory that can account for this behaviour and is therefore a useful tool for engineers when analysing deep beams.

A significant issue for Timoshenko finite element beam models is shear locking, in which finite elements exhibit increased stiffness in bending dominated problems [4]. For beams with high length to depth ratios, the influence of shear deformations become negligible, i.e. $\gamma_{xy} = 0$. The shear stress for a Timoshenko beam is derived in Chapter 4 and is given by:

$$\gamma_{xy} = v' - \theta \tag{2.1}$$

When both v and θ are approximated by linear functions, this can be rewritten as:

$$\gamma_{xy} = \left(\frac{v_R - v_L}{L} - \theta_L\right) - \frac{\theta_R - \theta_L}{L} x \tag{2.2}$$

To achieve zero shear strain for any value of x, both terms of equation 2.2 must be equal to zero independently of x. This means that:

$$\frac{v_R - v_L}{L} - \theta_L = 0 \quad \text{and} \quad \frac{\theta_R - \theta_L}{L} = 0 \tag{2.3a-b}$$

As a result, θ_R and θ_L must be equal, thus the polynomial that describes the rotation becomes constant. Since curvature is equal to the derivative of the rotation, then the curvature is equal to zero. This restricts any deformations of the element, leading to a stiffer response. If this problem is not avoided, the displacement of the element being analysed will be significantly underestimated, which would be unsafe for design.

There have been many methods developed over the decades to deal with shear locking. In the 1970s, the reduced integration technique was proposed that can eliminate the locking issue by reducing the number of integration points. The integration that is performed is one order below what is usually used. However, this can reduce the accuracy of the result obtained and instead lead to hourglassing, a phenomenon in which the elements become too flexible [6]. Bathe et al. proposed a technique in which the shear strain is calculated from separate interpolation functions rather than from the displacements [7]. However, this method is complicated because it is difficult to select the order of the interpolation functions [6]. Raveendranath et al. created a shear-flexible beam element that does not experience shear locking. To achieve this, force-moment and moment-shear equilibrium equations are solved simultaneously to derive the polynomial functions used to approximate the displacements [8]. However, this method has been found to reduce the accuracy of the finite elements [6].

For this thesis, a simple technique will be used to eliminate the issue of shear locking. From Equations 2.2 and 2.3, the issue of shear locking arose because both v and θ were approximated by linear functions, thus v' was a constant term and θ was a linear function. As a result, they did not have a compatible contributions. Instead, it should be ensured that for every nonzero displacement, the polynomials approximating the terms are of the same order. For example, in Equation 2.1, v' and θ must be of the same order, thus, the polynomial used to describe v would need to be of one order higher than the polynomial used to describe θ.

2.4: Generalised Beam Theory

There are some members for which in-plane effects cannot be ignored such as thin-walled members (TWM). Euler-Bernoulli beam theory and Timoshenko beam theory are not appropriate choices for the analysis of TWMs as they do not account for the effects of local or distortional buckling. In this case, it would be appropriate to use a more complicated beam theory to capture the behaviour of the TWM more accurately. In the mid-1960s, Richard Shardt introduced the Generalized Beam Theory (GBT). This theory can account for cross-section distortions, allowing for an effective model of thin-walled members to be obtained. This is achieved through its consideration of a members displacement field as a linear combination of the in plane and out of plane modes of the cross section and its intensity functions.

Richard Shardt first introduced GBT in the 1960s in the context of conducting analysis of prismatic thin-walled members. At this time, GBT considered only the conventional deformation modes. Shardt was largely responsible for the development of GBT during the final decades of the 20th century [9] Davies extended the use of GBT to investigate the buckling behaviour of cold formed steel members. However, there were still limitations to GBT as the contributions of Shardt and Davies only considered unbranched thin wall members made of linear elastic and isotropic materials [9].

Between 2000 and 2010, the applicability of GBT was further increased due to studies completed by a research team at the Technical University of Lisbon consisting of Camotim and co-workers. GBT was extended to handle arbitrarily branched open sections, nonlinear and orthotropic members [9]. Additionally, post-buckling, vibration and dynamic analysis was developed for both open and closed cross sections.

In recent years, research has been published that explores the use of GBT for new applications. Duan et al. extended GBT to apply to thin-walled members with holes of arbitrary shapes and placements [10]. As a result of this study, there is potential to increase the ease of assembly or enhance functionality of TWMs used in construction. Another recent development is the application of GBT to composite materials such as an arbitrary steel-concrete composite bridge deck and a composite beam with a thin-walled steel section [11][12]. This is significant due to the popularity of composite members in the building industry.

Chapter 3: Euler-Bernoulli Beam Theory

This chapter will outline the derivation of the finite elements for a Euler-Bernoulli beam. There are three key steps to developing a finite element beam model. Firstly, the governing differential equations that describe the behaviour of the beam must be derived. These equations, as well as the boundary conditions are known as the strong form. It is called the strong form because it requires that the solution must be continuous along the beam. This requirement makes the strong form difficult to solve because discontinuities arise when the geometry or material properties of the beam vary along its length. Therefore, the strong form must be transformed to an integral form of these equations, which is known as the weak form. The weak form provides a relaxation of the continuity requirements of the strong form and allows lower degree polynomials to be used for the solution. For this thesis, the weak form will be derived using the principle of virtual work. The final and most crucial step is to use polynomials to approximate the unknown functions. These approximation functions along with the weak form are used to derive the finite elements.

3.1: Assumptions

The derivation of the Euler-Bernoulli beam theory presented in this chapter is based on the following assumptions:

- Plane sections remain plane and normal to the member axis after deformation
- Deformed beam angles are small
- The beam has no imperfections and is initially straight
- The cross section does not deform significantly under the applied loading and can be assumed to be rigid
- The beam is symmetrical about the plane of bending
- The beam has a constant cross section along its length
- The beam is made of an isotropic and homogeneous material
- The stress levels are below the yield stress

Under these assumptions, no torsional or out of plane effects are considered. These assumptions are reasonable for beams with a large length to depth ratio, as these beams are dominated by flexural deformations.

It is important that the applied stress levels are below the yield. If an engineer was to use a linear Euler-Bernoulli beam model to investigate the behaviour of a beam that has yielded, they would obtain a result that does not reflect the real beam behaviour. This is because the stress in the beam is calculated using Hooke's law, which states that:

$$\sigma = E\varepsilon \tag{3.1}$$

Hooke's law is only valid in the linear elastic region. However, stress values will continue to be output for any given strain value. Therefore, it is crucial that a design engineer ensures that the stresses developed in the beam do not exceed the yield stress.

3.2: Strong Form

The strong form of EBBT is obtained using the kinematic model, which defines all possible displacements and deformations of the Euler-Bernoulli beam. The displacement field for a Euler-Bernoulli beam is presented in Figure 3.1, which reflects the assumption that plane sections remain plane and perpendicular to the member axis after deformation.

From Figure 3.1, the horizontal and vertical displacements of point 1 on the member axis (at an arbitrary level) due to the deformation are $u(x)$ and $v(x)$. The displacements at point 2 (not on the member axis) can be expressed in terms the displacements at point 1 as follows:

$$d_x(x, y) = u(x) - y_2 sin\theta \tag{3.2}$$

$$d_y(x, y) = v(x) + y_2 cos\theta - y_2 \tag{3.3}$$

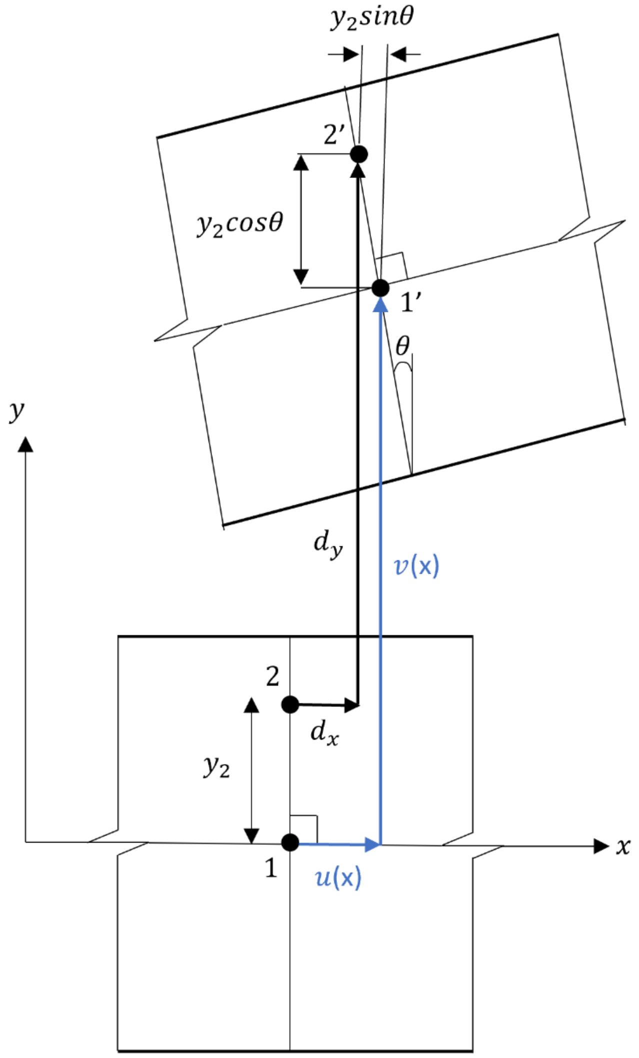


Figure 3.1: Displacement field for the Euler-Bernoulli Beam Model

For an arbitrary point on the cross section, by enforcing that displacements are small ($cos\theta \approx 1$ and $sin\theta \approx \theta$) and that plane sections remain plane and perpendicular to the member axis after deformation ($\theta = v'$) the displacements can be expressed as:

$$d_x(x,y) = u(x) - yv(x)' \tag{3.4}$$

$$d_y(x,y) = v(x) \tag{3.5}$$

$$d_z(x,y) = 0 \tag{3.6}$$

This results in the following strain field:

$$\varepsilon_x = \frac{\partial dx}{\partial x} = u(x)' - yv(x)'' \qquad \varepsilon_y = \frac{\partial dy}{\partial y} = 0 \qquad \varepsilon_z = \frac{\partial dz}{\partial z} = 0 \tag{3.7a-c}$$

$$\gamma_{xy} = \frac{\partial dx}{\partial y} + \frac{\partial dy}{\partial x} = 0 \qquad \gamma_{yz} = \frac{\partial dy}{\partial z} + \frac{\partial dz}{\partial y} = 0 \qquad \gamma_{xz} = \frac{\partial dx}{\partial z} + \frac{\partial dz}{\partial x} = 0 \tag{3.8a-c}$$

It can be seen from equations 3.7 and 3.8 that the only non-zero strain is ε_x, which is the strain normal to the cross-section in the x direction. The strain field also highlights that shear strain is not accounted for in the Euler-Bernoulli beam theory as all shear stresses are equal to zero.

3.3 Weak Form

The principal of virtual work is used to obtain the weak form of EBBT. An assumption of this principal is that the virtual work associated with the internal actions is equal to the virtual work done by the external actions. Figure 3.2 shows the free diagram of the beam segment under consideration. The internal actions consist of the longitudinal stress σ_x as well as its associated deformations. The external actions are presented in Figure 3.2 and consist of the member loads w and n as well as the nodal axial force (N), shear force (S), and moment (M) at each end of the beam.

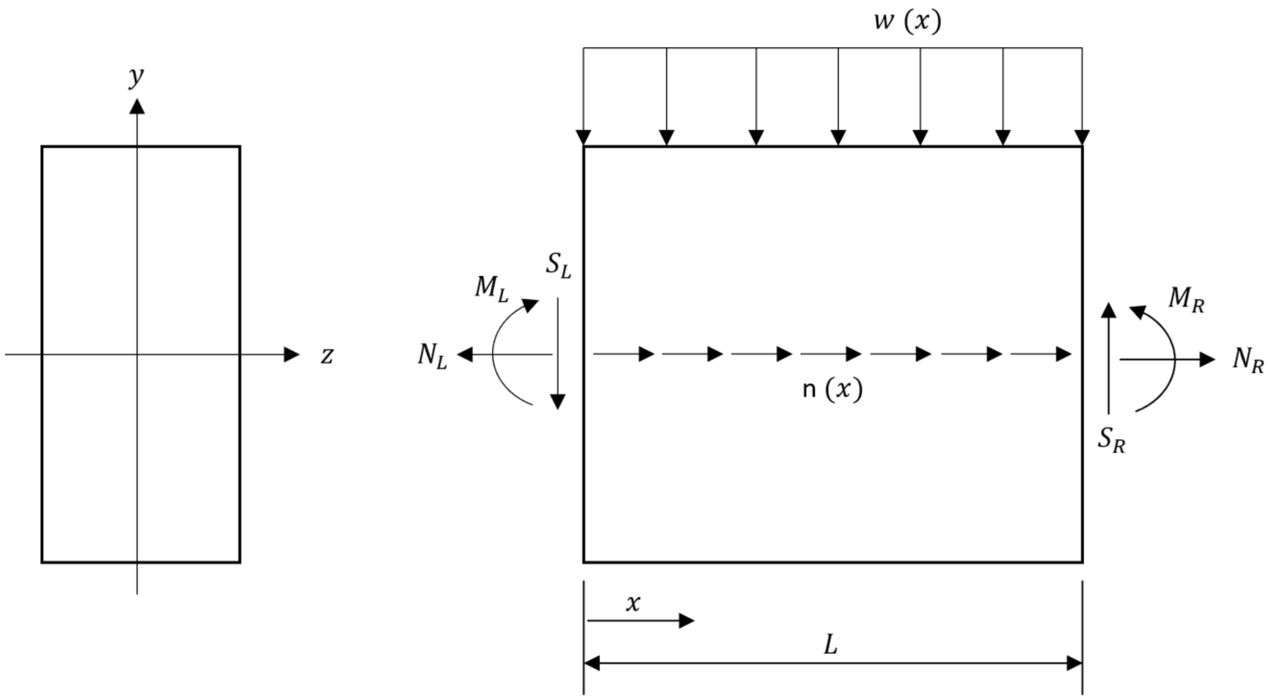


Figure 3.2: Free body diagram showing the member loads and nodal actions

To be able to calculate the internal forces along the beam, virtual displacements are used, which means that the choice of displacements are arbitrary given that they are consistent with the limitations of the structure under consideration. Equating the internal and external work gives the following expression:

$$\int_L \int_A \sigma_x \hat{\varepsilon}_x \, dA \, dx \;=\; \int_L (w\hat{v} \;+ n\hat{u}) \, dx \;-\; S_L \hat{v}_L - N_L \hat{u}_L - M_L \hat{\theta}_L + S_R \hat{v}_R + N_R \hat{u}_R + M_R \hat{\theta}_R \qquad (3.9)$$

where the subscripts L and R specify the end of the member that the load is acting and the hat '^' represents a virtual displacement or strain.

It is possible to assume that the end displacements have zero variation without affecting the derivation of the stiffness matrix and loading vectors for the finite elements. This allows equation 3.9 to be simplified to:

$$\int_L \int_A \sigma_x \hat{\varepsilon}_x \, dA \, dx \;=\; \int_L (w\hat{v} \;+ n\hat{u}) \, dx \qquad (3.10)$$

The internal axial force, N and moment, M have the following definitions and can be rewritten in terms of the geometric and material properties of the beam as follows:

$$N = \int_A \sigma_x \, dA \;=\; EA\varepsilon_r - EB\kappa \qquad (3.11)$$

$$M \;=\; -\int_A y\sigma_x \, dA \;=\; -EB\varepsilon_r - EI\kappa \qquad (3.12)$$

where E is the modulus of elasticity, B is the first moment of area, I is the second moment of area, ε_r is the strain at the reference axis and κ is the curvature.

By incorporating the definitions of the internal axial force N and moment M and substituting in Equation 3.7a for the axial strain, Equation 3.10 can be rewritten as:

$$\int_L \begin{bmatrix} EA & -EB \\ -EB & EI \end{bmatrix} \begin{bmatrix} \varepsilon_r \\ \kappa \end{bmatrix} \cdot \begin{bmatrix} \hat{u}' \\ \hat{v}'' \end{bmatrix} dx \;=\; \int_L \begin{bmatrix} n \\ w \end{bmatrix} \cdot \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix} dx \qquad (3.13)$$

which can be written in compact form as follows:

$$\int_L \boldsymbol{D}\hat{\boldsymbol{\varepsilon}} \cdot \boldsymbol{A}\hat{\boldsymbol{e}} \, dx \;=\; \int_L \boldsymbol{p} \cdot \hat{\boldsymbol{e}} \, dx \qquad (3.14)$$

where p is the vector of internal actions, D is a matrix containing the geometric and material properties of the cross section, A is a differential operator and e contains the generalised displacements u and v. These matrixes and vectors can be found in Appendix A.

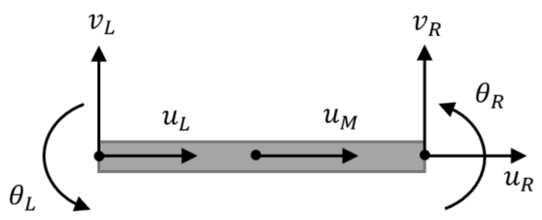
3.4 Finite Element Formulation

Now that the weak form of the EBBT has been derived, polynomial functions can be used to approximate the beam displacements. This is the key step to the development of the finite element model. To satisfy consistency requirements, the polynomials used to approximate the displacements must be chosen such that the terms within the expressions are of the same order. From Equation 3.6a, the non-zero strain for the Euler-Bernoulli beam model is given by:

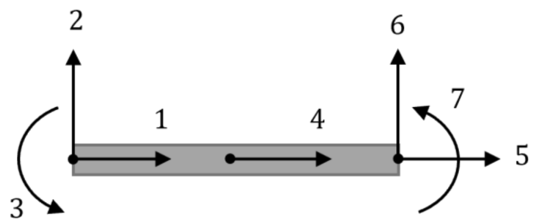
$$\varepsilon_x = u' - yv'' \tag{3.15}$$

Therefore, to achieve consistency, u' and v'' must be of the same order. This means that v must be approximated by a polynomial that is one order higher than the polynomial used to approximate u. The simplest element for which consistency is achieved is a seven degree of freedom (DOF) element, with the central degree of freedom being horizontal. This is demonstrated in appendix B.

Degrees of freedom are the possible movements or rotations that describe the deformation at a particular point of a beam. For the 2D beams explored in this thesis, each of the nodes have three freedoms. As presented in Figure 3.3, these are movements in the horizontal and vertical directions as well as a rotation. Each element also features a horizontal degree of freedom at its centre.



(a)



(b)

Figure 3.3: 7 DOF finite element for a Euler-Bernoulli beam. (a) Nodal displacements (b) Freedom numbering

For the seven DOF element presented in figure X, there are three horizontal DOFs, meaning that the axial displacement will be approximated by a parabolic function:

$$u = a_o + a_1 x + a_2 x^2 \tag{3.16}$$

Since there are four DOFs relating to the deflection, the vertical displacements will be approximated by a cubic polynomial:

$$v = b_0 + b_1 x + b_2 x^2 + b_3 x^3 \tag{3.17}$$

For finite element formulation, it is useful to express the coefficients a_i and b_i with terms that describe the displacements at the locations of DOFs. This is because it creates a direct connection between the adjacent finite elements. For axial displacement, this would be the axial displacement at the left node u_L, at the centre of the element u_M and at the right node u_R.

At the left node, x = 0, therefore:

$$u_L = a_o + (a_1 \times 0) + (a_2 \times 0^2) \tag{3.18}$$

At the centre of the element, x = $\frac{L}{2}$, which gives:

$$u_M = a_o + \left(a_1 \times \frac{L}{2}\right) + \left(a_2 \times \left(\frac{L}{2}\right)^2\right) \tag{3.19}$$

At the right node, x = L, therefore:

$$u_R = a_o + (a_1 \times L) + (a_2 \times L^2) \tag{3.20}$$

Therefore, the equations a_i can be rewritten in terms of u_L, u_M and u_R as:

$$a_o = u_L \qquad a_1 = -\frac{3u_L - 4u_M + u_R}{L} \qquad a_2 = \frac{2(u_L - 2u_M + u_R)}{L^2} \tag{3.21a-c}$$

Substituting Equations 3.21a-c into Equation 3.16 and isolating the displacements at each degree of freedom gives the following expression for the horizontal displacement:

$$u = N_{u1}u_L + N_{u2}u_M + N_{u3}u_R \tag{3.22}$$

where:

$$N_{u1} = 1 - \frac{3x}{L} + \frac{2x^2}{L^2} \qquad N_{u2} = \frac{4x}{L} - \frac{4x^2}{L^2} \qquad N_{u3} = -\frac{x}{L} + \frac{2x^2}{L^2} \tag{3.23a-c}$$

This process can be repeated for the deflection to obtain:

$$v = N_{v1}v_L + N_{v2}\theta_L + N_{v3}v_R + N_{v4}\theta_R \tag{3.24}$$

where:

$$N_{v1} = 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3} \qquad N_{v2} = x - \frac{2x^2}{L} + \frac{x^3}{L^2} \qquad N_{v3} = \frac{3x^2}{L^2} - \frac{2x^3}{L^3} \qquad N_{v4} = -\frac{x^2}{L} + \frac{x^3}{L^2} \tag{3.25a-d}$$

Equations 3.23 and 3.25 are known as shape functions as they describe the shape associated with each nodal displacement. Shape functions are used to interpolate the behaviour between the points of known displacement, which are the locations of the DOFs. These shape functions are equal to 1 at the location of their corresponding DOF and zero at the location of the other DOFs as presented in Figure 3.4.

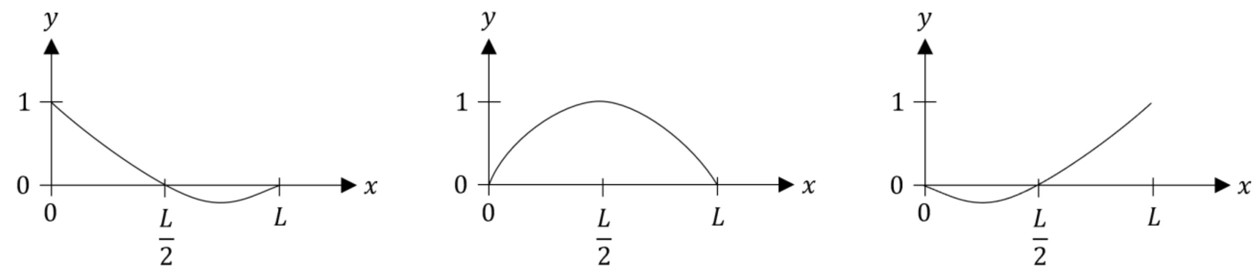


Figure 3.4: Shape functions for the second order approximation of u

Now that the displacements have been approximated by polynomial functions, the expressions for u and v can be input into the weak form given in Equation 3.14 to derive the element stiffness matrix and loading vector. This process is presented in Appendix A and gives:

$$k_e = \begin{bmatrix} \frac{7EA}{3L} & \frac{-4\mathrm{EB}}{L^2} & \frac{-3\mathrm{EB}}{L} & \frac{-8EA}{3L} & \frac{EA}{3L} & \frac{4\mathrm{EB}}{L^2} & \frac{-\mathrm{EB}}{L} \\ \frac{-4\mathrm{EB}}{L^2} & \frac{12EI}{L^3} & \frac{6EI}{L^2} & \frac{8\mathrm{EB}}{L^2} & \frac{-4\mathrm{EB}}{L^2} & \frac{-12EI}{L^3} & \frac{6EI}{L^2} \\ \frac{-3\mathrm{EB}}{L} & \frac{6EI}{L^2} & \frac{4EI}{L} & \frac{4\mathrm{EB}}{L} & \frac{-\mathrm{EB}}{L} & \frac{-6EI}{L^2} & \frac{2EI}{L} \\ \frac{-8EA}{3L} & \frac{8\mathrm{EB}}{L^2} & \frac{4\mathrm{EB}}{L} & \frac{16EA}{3L} & \frac{-8EA}{3L} & \frac{-8\mathrm{EB}}{L^2} & \frac{4\mathrm{EB}}{L} \\ \frac{EA}{3L} & \frac{-4\mathrm{EB}}{L^2} & \frac{-\mathrm{EB}}{L} & \frac{-8EA}{3L} & \frac{7EA}{3L} & \frac{4\mathrm{EB}}{L^2} & \frac{-3\mathrm{EB}}{L} \\ \frac{4\mathrm{EB}}{L^2} & \frac{-12EI}{L^3} & \frac{-6EI}{L^2} & \frac{-8\mathrm{EB}}{L^2} & \frac{4\mathrm{EB}}{L^2} & \frac{12EI}{L^3} & \frac{-6EI}{L^2} \\ \frac{-\mathrm{EB}}{L} & \frac{6EI}{L^2} & \frac{2EI}{L} & \frac{4\mathrm{EB}}{L} & \frac{-3\mathrm{EB}}{L} & \frac{-6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \tag{3.26}$$

$$q_e = \begin{bmatrix} \frac{L}{6}n & \frac{L}{2}w & \frac{L^2}{12}w & \frac{2L}{3}n & \frac{L}{6}n & \frac{L}{2}w & \frac{-L^2}{12}w \end{bmatrix}^T \tag{3.27}$$

where E, A, L and B denote the modulus of elasticity, cross sectional area, length of the finite element and first moment of area respectively.

The contributions of each element must be collected to form a global stiffness matrix. To achieve this, the load and displacement vectors for each element, which may have different orientations, must first be expressed in global coordinates.

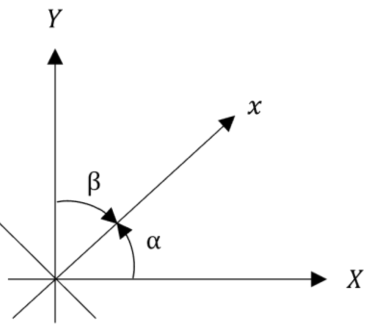


Figure 3.5: Angles relating local and global coordinate axes.

There are two types of coordinate systems used for this finite element model. The first is the local coordinate system (x, y), which is related to an individual element of the beam. The local x-axis is parallel to the member and the local y-axis is perpendicular to it. However, because the elements of the beam may not be oriented in the same direction, the local coordinate system of one element may not be the applicable to all the other elements. For this reason, a global coordinate system (X, Y) that is applicable to the entire member is necessary.

Figure 3.5 presents the angles which relate the local and global coordinate systems. The transformation matrix T, uses the cosines of these angles to convert the local coordinate axis to the global coordinate axis and is given by:

$$T = \begin{bmatrix} l & m & 0 & 0 & 0 & 0 & 0 \\ -m & l & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & l & m & 0 \\ 0 & 0 & 0 & 0 & -m & l & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.28}$$

where:

$$l = \cos\alpha \qquad m = \cos\beta \tag{3.29}$$

This transformation matrix is used to convert the components of the displacement and loading vectors into global components using the following expressions:

$$\boldsymbol{d}_e = \boldsymbol{T}\boldsymbol{D}_e \qquad \boldsymbol{Q}_e = \boldsymbol{T}^T\boldsymbol{q} \qquad \boldsymbol{K}_e = \boldsymbol{T}^T\boldsymbol{k}_e\boldsymbol{T} \tag{3.30a-c}$$

The contribution of each element can then be assembled to produce the stiffness relationship of the entire structure using the following relationship:

$$\boldsymbol{Q} = \boldsymbol{K}\boldsymbol{D} \tag{3.31}$$

where Q is the vector of the nodal actions of the entire structure, K is the global stiffness matrix and D contains the displacements for the whole structure. The Q vector currently only accounts for the loads due to the distributed member loading, however, any point loads should be added at the corresponding degrees of freedom before proceeding.

The Q and D vectors will both have sizes equal to the number of global freedoms, N_{DOF}. The global stiffness matrix will be a $N_{DOF} \times N_{DOF}$ square matrix. For the seven DOF element presented in Figure 3.3, the total number of degrees of freedom in the member is related to the number of elements ($N_{Elements}$) in the following fashion:

$$N_{DOF} = (4 \times N_{Elements}) + 3 \tag{3.32}$$

Equation 3.31 can be rewritten to separate the unknown and known displacements and actions as follows:

$$\begin{bmatrix} \boldsymbol{Q}_k \\ \boldsymbol{Q}_u \end{bmatrix} = \begin{bmatrix} \boldsymbol{K}_{11} & \boldsymbol{K}_{12} \\ \boldsymbol{K}_{21} & \boldsymbol{K}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{D}_u \\ \boldsymbol{D}_k \end{bmatrix} \tag{3.33}$$

The unknown displacements and actions can then be calculated using:

$$\boldsymbol{D}_u = \boldsymbol{K}_{11}^{-1}(\boldsymbol{Q}_k - \boldsymbol{K}_{12}\boldsymbol{D}_k) \qquad \boldsymbol{Q}_{\boldsymbol{u}} = \boldsymbol{K}_{21}\boldsymbol{D}_{\boldsymbol{u}} + \boldsymbol{K}_{22}\boldsymbol{D}_k \tag{3.34a-b}$$

3.5: Example Calculation

The following example will demonstrate the process for producing and partitioning the stiffness matrix K into the form required for Equations 3.33 and 3.34. Figure 3.6 presents the simply supported beam under consideration. The beam has a square cross section and has the following properties: b (width) and d (depth) = 200 mm and E = 20000 MPa. A centroidal reference system is adopted such that B = 0.

The restrained freedoms are coloured in red while the unrestrained freedoms are coloured in black. Because the beam is simply supported, the left node will have its horizontal and vertical freedoms restrained while the right node will have only its vertical freedom restrained. This corresponds with freedoms 1, 2 and 10.

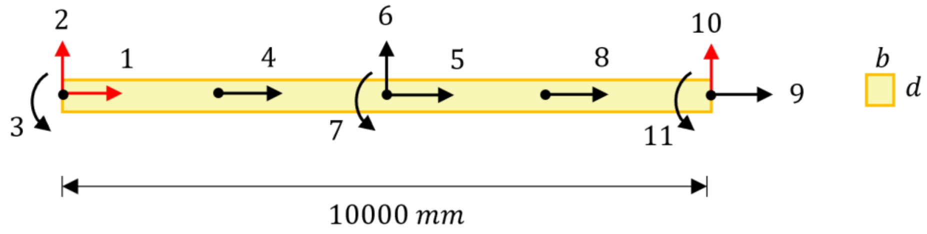


Figure 3.6: Beam for example calculation

For ease of calculation, the beam is discretised into two elements with seven degrees of freedom each as shown in Figure 3.7. These elements have the same cross section, length, and material properties. Additionally, they are oriented such that their local axis is coincident with the global axis (i.e. the local element stiffness matrix is equal to the global element stiffness matrix). Therefore, both elements have the same global element stiffness matrix which is calculated using Equation 3.15 and presented in Figure 3.8.

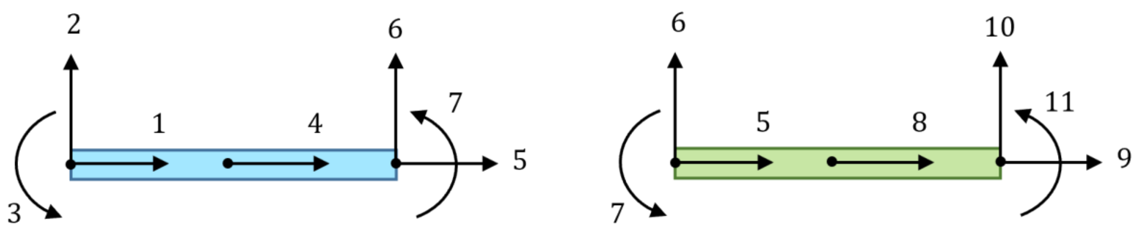


Figure 3.7: Finite elements for example calculation

Although the element stiffness matrices are identical for both elements, the degrees of freedom associated with each entry of the matrix varies for each element. This is illustrated in Figure 3.8 where the blue box containing numbers 1-7 indicates the corresponding freedoms for the first element while the green box containing numbers 5-11 indicates the corresponding freedoms for the second element.

The elements have one shared node and therefore share three freedoms, which are freedoms 5, 6 and 7. When the elemental stiffness matrices are consolidated into the global stiffness matrix, the contributions of each of the shared freedoms must be added together.

$$\begin{array}{ccc}
 & \begin{array}{ccccccc} 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} & \\
K_e = 10^3 \times & \begin{bmatrix}
373.3 & 0 & 0 & -426.7 & 53.33 & 0 & 0 \\
0 & 0.256 & 640 & 0 & 0 & -0.256 & 640 \\
0 & 640 & 2133333 & 0 & 0 & -640 & 1066667 \\
-426.7 & 0 & 0 & 853.3 & -426.7 & 0 & 0 \\
53.33 & 0 & 0 & -426.7 & 373.3 & 0 & 0 \\
0 & -0.256 & -640 & 0 & 0 & 0.256 & -640 \\
0 & 640 & 1066667 & 0 & 0 & -640 & 2133333
\end{bmatrix} & \begin{array}{cc} 1 & 5 \\ 2 & 6 \\ 3 & 7 \\ 4 & 8 \\ 5 & 9 \\ 6 & 10 \\ 7 & 11 \end{array}
\end{array}$$

Figure 3.8: Element stiffness matrix for example calculation

For example, the entries with the 5th degree of freedom as its row and column index are highlighted for each element in Figure 3.8. Both have a value of 373.33×10^3 which when added together give a value of 746.6×10^3. This is identical to the corresponding entry for the global stiffness matrix with the fifth degree of freedom as its row and column index which is highlighted yellow in Figure 3.9.

$$K = 10^3 \times \begin{array}{cc} \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \end{matrix} & \\ \begin{bmatrix} 373.3 & 0 & 0 & -426.7 & 53.33 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.256 & 640 & 0 & 0 & -0.256 & 640 & 0 & 0 & 0 & 0 \\ 0 & 640 & 2133333 & 0 & 0 & -640 & 1066667 & 0 & 0 & 0 & 0 \\ -426.7 & 0 & 0 & 853.3 & -426.7 & 0 & 0 & 0 & 0 & 0 & 0 \\ 53.33 & 0 & 0 & -426.7 & 746.7 & 0 & 0 & -425.7 & 53.33 & 0 & 0 \\ 0 & -0.256 & -640 & 0 & 0 & 0.512 & 0 & 0 & 0 & -0.256 & 640 \\ 0 & 640 & 1066667 & 0 & 0 & 0 & 4266667 & 0 & 0 & -640 & 1066667 \\ 0 & 0 & 0 & 0 & -426.7 & 0 & 0 & 853.3 & -426.7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 53.33 & 0 & 0 & -426.7 & 373.3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -0.256 & -640 & 0 & 0 & 0.256 & -640 \\ 0 & 0 & 0 & 0 & 0 & 640 & 1066667 & 0 & 0 & -640 & 2133333 \end{bmatrix} & \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \end{matrix} \end{array}$$

Figure 3.9: Global stiffness matrix for example calculation

Once the global stiffness matrix has been assembled, it must be partitioned to allow the unknown loads and displacements to be calculated using Equations 3.34. Ideally, when computing a hand calculation, the stiffness matrix would be assembled such that the restrained freedoms are listed first followed by the unrestrained freedoms. This is because it allows for much more convenient partitioning. However, this is inefficient for the large number of freedoms that are used in finite element modelling. Therefore, the freedoms associated with the restrained (1, 2 and 10) and unconstrained freedoms (3-9 and 11) are instead recorded separately and used to partition the matrix into four sub-matrices as presented in Figure 3.10 with K_{11} highlighted in purple, K_{12} highlighted in green, K_{21} highlighted in blue and K_{22} highlighted in yellow.

$$K = 10^3 \times \begin{array}{cc} \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \end{matrix} & \\ \begin{bmatrix} 373.3 & 0 & 0 & -426.7 & 53.33 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.256 & 640 & 0 & 0 & -0.256 & 640 & 0 & 0 & 0 & 0 \\ 0 & 640 & 2133333 & 0 & 0 & -640 & 1066667 & 0 & 0 & 0 & 0 \\ -426.7 & 0 & 0 & 853.3 & -426.7 & 0 & 0 & 0 & 0 & 0 & 0 \\ 53.33 & 0 & 0 & -426.7 & 746.7 & 0 & 0 & -425.7 & 53.33 & 0 & 0 \\ 0 & -0.256 & -640 & 0 & 0 & 0.512 & 0 & 0 & 0 & -0.256 & 640 \\ 0 & 640 & 1066667 & 0 & 0 & 0 & 4266667 & 0 & 0 & -640 & 1066667 \\ 0 & 0 & 0 & 0 & -426.7 & 0 & 0 & 853.3 & -426.7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 53.33 & 0 & 0 & -426.7 & 373.3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -0.256 & -640 & 0 & 0 & 0.256 & -640 \\ 0 & 0 & 0 & 0 & 0 & 640 & 1066667 & 0 & 0 & -640 & 2133333 \end{bmatrix} & \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \end{matrix} \end{array}$$

Figure 3.10: Partitioned global stiffness matrix for example calculation

K_{11} contains the entries of the global stiffness matrix that have a restrained freedom number for both its row and column indices and is given by:

$$K_{11} = 10^3 \times \begin{bmatrix} 373.3 & 0 & 0 \\ 0 & 0.256 & 0 \\ 0 & 0 & 0.256 \end{bmatrix} \tag{3.35}$$

K_{12} contains the entries of the global stiffness matrix that have an unrestrained freedom for its column index and a restrained freedom as its row index and is given by:

$$K_{12} = 10^3 \times \begin{bmatrix} 0 & -426.7 & 53.33 & 0 & 0 & 0 & 0 & 0 \\ 640 & 0 & 0 & -0.256 & 640 & 0 & 0 & 0 \\ 0 & 0 & 0 & -0.256 & -640 & 0 & 0 & -640 \end{bmatrix} \tag{3.36}$$

K_{21} contains the entries of the global stiffness matrix with a restrained freedom for its column index and an unrestrained freedom as its row index. For this scenario, K_{21} is given by:

$$K_{21} = 10^3 \times \begin{bmatrix} 0 & 640 & 0 \\ -426.7 & 0 & 0 \\ 53.33 & 0 & 0 \\ 0 & -0.256 & -0.256 \\ 0 & 640 & -640 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -640 \end{bmatrix} \tag{3.37}$$

Finally, K_{22} contains the entries of the global stiffness matrix that have an unrestrained freedom number for its row and column indices and is given by:

$$K_{22} = 10^3 \times \begin{bmatrix} 2133333 & 0 & 0 & -640 & 1066667 & 0 & 0 & 0 \\ 0 & 853.3 & -426.7 & 0 & 0 & 0 & 0 & 0 \\ 0 & -426.7 & 746.7 & 0 & 0 & -425.7 & 53.33 & 0 \\ -640 & 0 & 0 & 0.512 & 0 & 0 & 0 & 640 \\ 1066667 & 0 & 0 & 0 & 4266667 & 0 & 0 & 1066667 \\ 0 & 0 & -426.7 & 0 & 0 & 853.3 & -426.7 & 0 \\ 0 & 0 & 53.33 & 0 & 0 & -426.7 & 373.3 & 0 \\ 0 & 0 & 0 & 640 & 1066667 & 0 & 0 & 2133333 \end{bmatrix} \tag{3.38}$$

A similar process should be followed when developing the Q and D vectors. Q contains the actions due to the applied loading as well as the reactions at the supports. When partitioning Q into Q_u and Q_k, Q_u refers to the unknown actions which are those at the restrained freedoms because the reactions at the supports must be determined during the analysis. Q_k contains the known actions, which are those at the unrestrained freedoms. When partitioning D into D_u and D_k, D_u refers to the unknown displacements, which are the displacements of the unrestrained freedoms as these are free to move. On the other hand, D_k refers to the displacements of the restrained freedoms as these are unable to move and are therefore equal to zero.

3.6: Post Processing

Once the unknown displacements have been calculated, the details of the structural response of the member can be calculated using the nodal displacements. This is highly useful to a design engineer as they can extract useful information about the beam behaviour such as the strain, normal force and bending moment at each point along the beam. For example, the strain at the reference axis and the curvature can be obtained from the following relationship:

$$\begin{bmatrix} \varepsilon_r \\ \kappa \end{bmatrix} = \begin{bmatrix} N'_{u1} & 0 & 0 & N'_{u2} & N'_{u3} & 0 & 0 \\ 0 & N''_{v1} & N''_{v2} & 0 & 0 & N''_{v3} & N''_{v4} \end{bmatrix} \begin{bmatrix} u_L \\ v_L \\ \theta_L \\ u_M \\ u_R \\ v_L \\ \theta_L \end{bmatrix} \tag{3.39}$$

which gives that:

$$\varepsilon_r = -\frac{3}{L}u_L + \frac{4}{L}u_M - \frac{1}{L}u_R + (\frac{4}{L^2}u_L - \frac{8}{L^2}u_M + \frac{4}{L^2}u_R)x \tag{3.40}$$

$$\kappa = -\frac{6}{L^2}v_L - \frac{4}{L}\theta_L + \frac{6}{L^2}v_R - \frac{2}{L}\theta_R + (\frac{12}{L^3}v_L + \frac{6}{L^2}\theta_L - \frac{12}{L^{33}}v_L + \frac{6}{L^2}\theta_R)x \tag{3.41}$$

The values obtained from Equations 3.40 and 3.41 can be substituted into Equations 3.11 and 3.12 to obtain the axial force and bending moment diagrams of the beam.

Chapter 4: Timoshenko Beam Theory

This chapter will outline the development of finite elements for a Timoshenko beam. The general process for deriving the finite elements using Timoshenko Beam Theory (TBT) is the same as for the Euler-Bernoulli Beam Theory. The difference between the two beam models is that while both EBBT and TBT assume that plane sections remain plane after deformations, EBBT assumes that they remain orthogonal to the member axis, while TBT does not. The significance of the relaxation of this Euler-Bernoulli assumption will be demonstrated in this chapter.

4.1: Assumptions

The derivation of the Timoshenko beam theory presented in this chapter is based on the following assumptions:

- Plane sections remain plane after deformation
- Deformed beam angles are small
- The beam initially has no imperfections and is initially straight
- The distribution of the shear stress over the cross section is constant
- The beam is symmetrical about the plane of bending
- The beam has a constant cross section along its length
- The beam is made of a homogeneous material
- The stress levels are below the yield stress

4.2: Strong Form

As for EBBT, the strong form of the governing equations of TBT is obtained from its kinematic model. From Figure 4.1, the cross section is no longer perpendicular to the member axis because of shearing deformations.

The initial equations describing the displacement of the beam are identical to those for EBBT:

$$d_x(x, y) = u(x) - y_2 sin\theta \quad (4.1)$$

$$d_y(x, y) = v(x) + y_2 cos\theta - y_2 \quad (4.2)$$

The Timoshenko Beam Model still assumes small displacements, i.e. $cos\theta \approx 1$ and $sin\theta \approx \theta$. However, due to the shearing deformations that are present in a Timoshenko beam, the rotation of the beam is not always equal to v'. This gives the following expressions for the generalised displacements:

$$d_x(x, y) = u(x) - y\theta \quad (4.3)$$

$$d_y(x, y) = v(x) \quad (4.4)$$

$$d_z(x, y) = 0 \quad (4.5)$$

Therefore, the strain field is given by:

$$\varepsilon_x = \frac{\partial dx}{\partial x} = u(x)' - y\theta(x)' \qquad \varepsilon_y = \frac{\partial dy}{\partial y} = 0 \qquad \varepsilon_z = \frac{\partial dz}{\partial z} = 0 \quad (4.6a\text{-}c)$$

$$\gamma_{xy} = \frac{\partial dx}{\partial y} + \frac{\partial dy}{\partial x} = -\theta + v' \qquad \gamma_{yz} = \frac{\partial dy}{\partial z} + \frac{\partial dz}{\partial y} = 0 \qquad \gamma_{xz} = \frac{\partial dx}{\partial z} + \frac{\partial dz}{\partial x} = 0 \quad (4.7a\text{-}c)$$

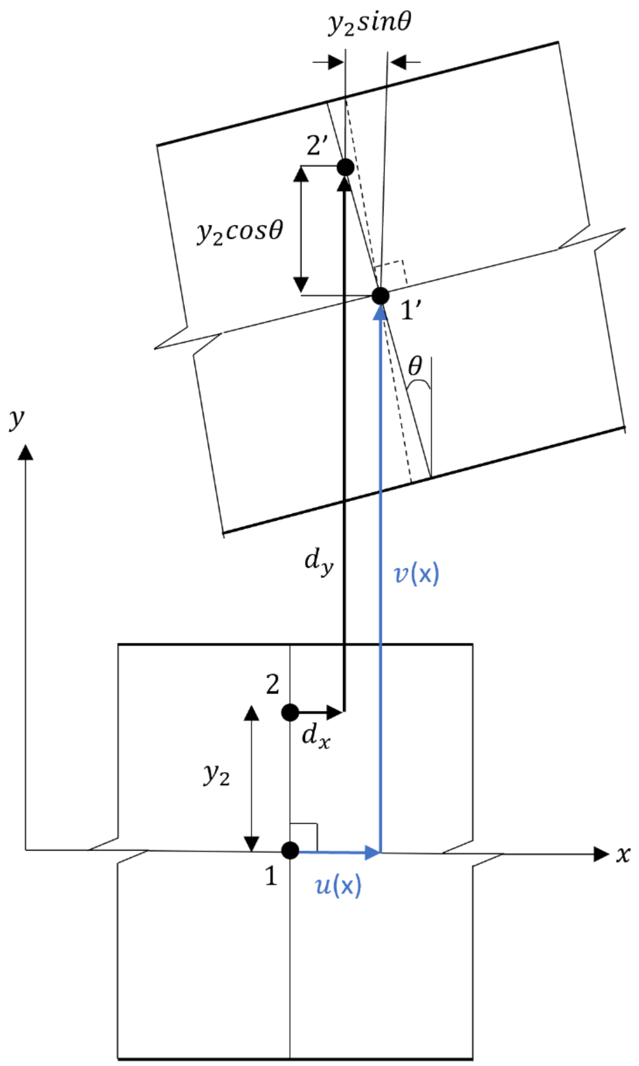
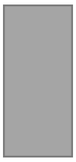
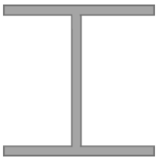
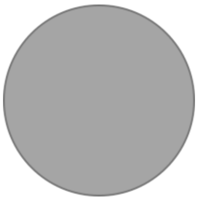
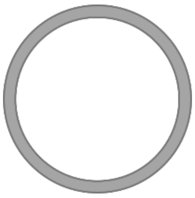


Figure 4.1: Displacement field for the Timoshenko Beam Model

It can be seen from Equations 4.6 and 4.7 that there are now two non-zero strains. These are the strain normal to the cross-section and the shear strain in the x direction. This highlights the capacity for TBT to account for shearing deformations.

TBT is a first order shear deformation theory and therefore has a constant shear strain for each cross section [13]. This is highlighted in Equation 4.7a, which shows that the shear stress is constant for every point of the cross section. This leads to inaccuracies as the top and bottom fibres of the cross section should be stress free to satisfy equilibrium and the distribution should be quadratic. To be able to capture the actual shear strain distribution along the cross section, a higher order polynomial would be necessary. This issue can be mitigated by using a shear correction factor when using TBT. For the rectangular cross sections investigated in this thesis, the shear correction factor (k_s) is equal to 5/6. Table 4.1 presents the shear correction factors for some common cross sections.

Table 4.1: Shear correction factor for some common cross-section shapes [4]

Diagram	k_s
	$\frac{5}{6}$
	Major axis: 0.69 Minor axis: 0.32
	$\frac{6}{7}$
	0.5
	0.416

4.3: Weak Form

As for EBBT, the weak form of TBT is derived using the principle of virtual work. The internal stresses include the longitudinal stresses σ_x and the shearing stresses τ_x as well as their associated deformations. The external stresses consist of the member loads w and n as well as the nodal actions at the left and right ends of the beam. This gives the following expression for the principle of virtual work:

$$\int_L \int_A \sigma_x \hat{\varepsilon}_x \, dA \, dx + \int_L \int_A \tau_x \hat{\gamma}_x \, dA \, dx = \int_L (w\hat{v} + n\hat{u}) \, dx + S_L \hat{v}_L + N_L \hat{u}_L + M_L \hat{\theta}_L + S_R \hat{v}_R + N_R \hat{u}_R + M_R \hat{\theta}_R \qquad (4.8)$$

The definitions of the internal axial force and moment are given in Equations 3.11 and 3.12. The internal shear force, S is defined as follows and can be expressed in terms of the geometric and material properties as:

$$S = \int_A \tau_{xy} \, dA = k_s A G \gamma_{xy} \tag{4.9}$$

where G is the shear modulus, γ_{xy} is the shear strain and k_s is the shear correction factor

Following the same procedure as for EBBT, Equation 4.8 can be rewritten as:

$$\int_L \begin{bmatrix} N \\ M \\ S \end{bmatrix} \cdot \begin{bmatrix} \hat{u}' \\ \hat{\theta}' \\ \hat{v}' - \hat{\theta} \end{bmatrix} dx = \int_L \begin{bmatrix} n \\ w \\ 0 \end{bmatrix} \cdot \begin{bmatrix} \hat{u} \\ \hat{v} \\ \hat{\theta} \end{bmatrix} dx \tag{4.10}$$

which can be written in compact form as follows:

$$\int_L \boldsymbol{D}\hat{\boldsymbol{\varepsilon}} \cdot \boldsymbol{A}\hat{\boldsymbol{e}} \, dx = \int_L \boldsymbol{p} \cdot \hat{\boldsymbol{e}} \, dx \tag{4.11}$$

This form is identical to what was developed for the EBBT, however the terms within the vectors and matrices are different. These matrixes and vectors can be found in Appendix C.

4.4: Finite Element Formulation

As discussed in Chapter 2, Timoshenko finite elements are susceptible to a phenomenon known as shear locking that causes the beam to exhibit stiffer behaviour than expected. This is dealt with by ensuring that the strain equations in Equations 4.6a and 4.7a feature terms of the same order. A seven degree of freedom element is used for a Timoshenko beam, as it is the simplest element that satisfies the consistency requirements and avoids shear locking problems. EBBT also used a seven DOF element, however, TBT requires that the central degree of freedom is vertical rather than horizontal as shown in Figure 4.2.

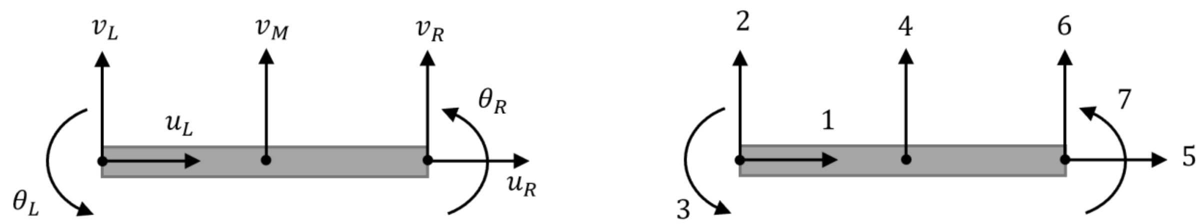


Figure 4.2: 7 DOF finite element for a Timoshenko beam. (a) Nodal displacements. (b) Freedom numbering

For the seven DOF element presented, there are two horizontal and rotational DOFs, meaning that the displacements u and θ will be approximated by a linear function. This means that u' and θ' will both be constants, allowing the terms of Equation 4.6a to be of the same order. Since there are three vertical DOFs, the vertical displacements will be approximated by a parabolic function. As a result, v' will be a first order term, which is equal to the order of θ, thus allowing equation 4.7a to be consistent. Hence, the finite element presented in Figure 4.2 will not be subjected to shear locking effects.

Following the same procedure as for EBBT, the displacement within each finite element can be expressed in terms of the displacement at the locations of the DOFs as follows:

$$u = N_{u1}u_L + N_{u2}u_R \qquad v = N_{v1}v_L + N_{v2}V_M + N_{v3}v_R \qquad \theta = N_{\theta 1}\theta_L + N_{\theta 2}\theta_R \tag{4.12a-c}$$

where:

$$N_{u1} = 1 - \frac{x}{L} \qquad N_{u2} = \frac{x}{L} \tag{4.13a-b}$$

$$N_{v1} = 1 - \frac{3x}{L} + \frac{2x^2}{L^2} \quad N_{v2} = \frac{4x}{L} - \frac{4x^2}{L^2} \qquad N_{v3} = \frac{2x^2}{L^2} - \frac{x}{L} \tag{4.14a-c}$$

$$N_{\theta 1} = 1 - \frac{x}{L} \qquad N_{\theta 2} = \frac{x}{L} \tag{4.15a-b}$$

Following the same procedure as described in Chapter 3 provides the following elemental stiffness matrix and loading vector:

$$\boldsymbol{k}_e = \begin{bmatrix} \frac{EA}{L} & 0 & \frac{-EB}{L} & 0 & \frac{-EA}{L} & 0 & \frac{EB}{L} \\ 0 & \frac{7k_sAG}{3L} & \frac{5k_sAG}{6} & \frac{-8k_sAG}{3L} & 0 & \frac{k_sAG}{3L} & \frac{k_sAG}{6} \\ \frac{-EB}{L} & \frac{5k_sAG}{6} & \frac{LAG}{3}+\frac{EI}{L} & \frac{-2k_sAG}{3} & \frac{EB}{L} & \frac{-k_sAG}{6} & \frac{Lk_sAG}{6}-\frac{EI}{L} \\ 0 & \frac{-8k_sAG}{3L} & \frac{-2k_sAG}{3} & \frac{16k_sAG}{3L} & 0 & \frac{-8k_sAG}{3L} & \frac{2k_sAG}{3} \\ \frac{-EA}{L} & 0 & \frac{EB}{L} & 0 & \frac{EA}{L} & 0 & \frac{-EB}{L} \\ 0 & \frac{k_sAG}{3L} & \frac{-k_sAG}{6} & \frac{-8k_sAG}{3L} & 0 & \frac{7k_sAG}{3L} & \frac{-5k_sAG}{6} \\ \frac{EB}{L} & \frac{k_sAG}{6} & \frac{Lk_sAG}{6}-\frac{EI}{L} & \frac{2k_sAG}{3} & \frac{-EB}{L} & \frac{-5k_sAG}{6} & \frac{Lk_sAG}{3}+\frac{EI}{L} \end{bmatrix} \tag{4.16}$$

$$\boldsymbol{q}_e = \begin{bmatrix} \frac{nL}{2} & \frac{wL}{6} & 0 & \frac{2wL}{3} & \frac{nL}{2} & \frac{wL}{6} & 0 \end{bmatrix}^T \tag{4.17}$$

Once the elemental stiffness matrix and loading vector have been obtained, the same procedure outlined in equations 3.28 to 3.34 are to be followed to solve for the unknown reactions and displacements along the beam.

4.5: Post Processing

As for EBBT, once the unknown displacements are calculated, the variables describing its structural response can be calculated along each finite element using the nodal displacements. The strain variables can be calculated using the following expression:

$$\begin{bmatrix} \varepsilon_r \\ \kappa \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} N'_{u1} & 0 & 0 & 0 & N'_{u2} & 0 & 0 \\ 0 & 0 & N'_{\theta 1} & 0 & 0 & 0 & N'_{\theta 2} \\ 0 & N'_{v1} & -N_{\theta 1} & N'_{v2} & 0 & N'_{v3} & N_{\theta 2} \end{bmatrix} \begin{bmatrix} u_L \\ v_L \\ \theta_L \\ v_M \\ u_R \\ v_L \\ \theta_L \end{bmatrix} \tag{4.18}$$

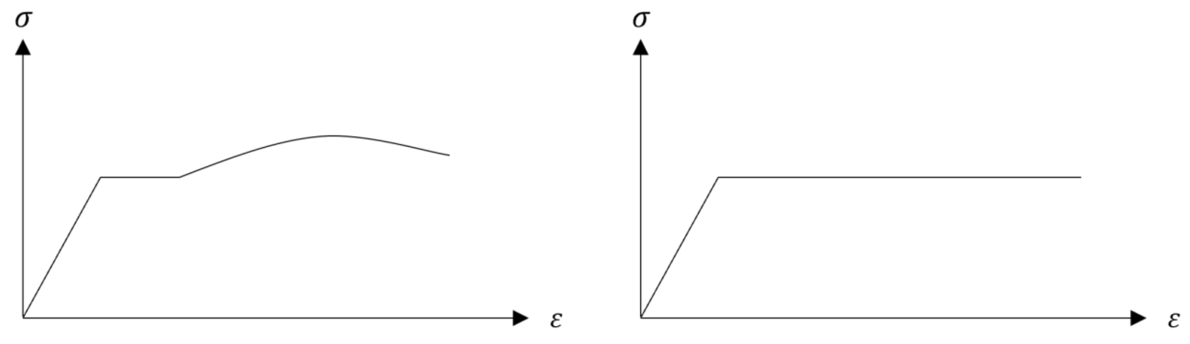
Chapter 5: Nonlinear Analysis

5.1: Background

This chapter will explore the nonlinear finite element analysis of a Euler Bernoulli beam using the Newton-Raphson method. There are many types of nonlinearity in structures, including geometric nonlinearity, contact nonlinearity and material nonlinearity. Geometric nonlinearity occurs because of large deformations that cause significant structural changes. This is uncommon for engineering problems as deformations are not permitted to reach such severe levels. As a result, most structural analysis assumes that the structure is geometrically linear to avoid unnecessary complexity. Therefore, it will not be considered in this thesis. Contact nonlinearity occurs due to a change in the boundary conditions of the structure. This can occur due to different elements pushing or pulling on each other. Since this is not in the scope of this thesis, it will not be explored.

Material nonlinearity occurs when the structure is loaded such that the stresses experienced are greater than the yield strength of the material. In this case, the stress strain relationship can no longer be expressed using Hooke's law ($\sigma = E\varepsilon$). Accounting for material nonlinearity is useful when evaluating a beam's behaviour close to its point of failure. This is the type of nonlinearity that will be explored in this thesis.

A constitutive model is an equation that describes the stress strain relationship of a material. One such equation is Hooke's law. The constitutive model that is chosen for analysis depends on the depth to which the material nonlinearity will be explored.



(a) (b)

Figure 5.1: Stress strain curves for a (a) ductile material (b) elastic-perfectly plastic material

In the case of mild steel, the stress strain behaviour will occur as shown in Figure 5.1a. However, for simplicity, the behaviour will be idealised in this thesis, utilising elastic-perfectly plastic behaviour as presented in Figure 5.1b. This means that for strain less than or equal to the yield strain, Hooke's law will be used to determine the stress in the beam. However, once the yield strain is surpassed, the stress will be assumed to be equal to the yield stress. In summary:

$$\sigma = E\varepsilon \quad \text{for } |\varepsilon| \leq \varepsilon_y \tag{5.1}$$

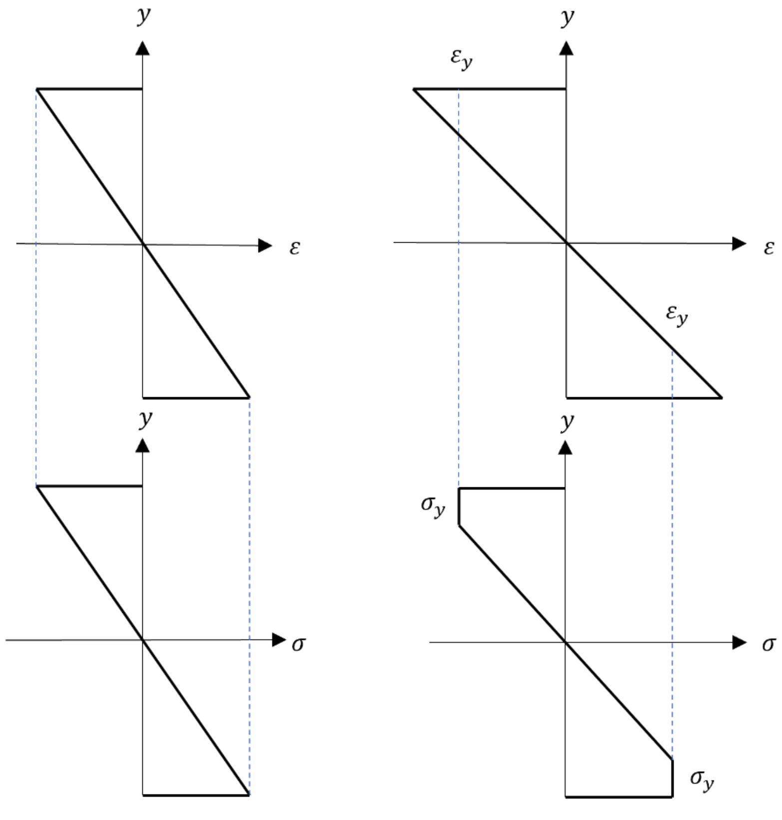
$$\sigma = f_y \quad \text{for } \varepsilon > \varepsilon_y \qquad\qquad \sigma = -f_y \quad \text{for } \varepsilon < -\varepsilon_y \tag{5.2a-b}$$

where:

$$\varepsilon_y = \frac{f_y}{E} \tag{5.3}$$

This idealisation reduces the accuracy of the obtained solution but is a conservative assumption. If a designer wanted to achieve a more economical design, they could use a more realistic constitutive model. For this analysis, it will be assumed that the material behaves equally in tension and compression. Therefore, this model will not be applicable to concrete due to its brittle behaviour and its weakness in tension. The model developed in this chapter will be best applied to ductile materials such as metal.

The constitutive model is represented visually in Figure 5.2. Figure 5.2a, shows the linear distribution of stress and strain within the linear elastic region. However, once the yield strain is exceeded, the stress at the extreme fibres remains equal to the yield strength of the material and the excess load begins to be redistributed toward the centre of the cross section as presented in Figure 5.2b. This will continue until the central fibre of the cross section reaches yield and the load cannot be further redistributed. This allows the entire capacity of the cross section to be utilised because as opposed to linear analysis where the central portion of the cross section will only experience extremely small stresses. Utilising the entire cross section will enable smaller members to be used, allowing a more cost-effective solution to be developed.



(a) (b)

Figure 5.2: Strain and stress diagrams for a cross section in (a) the linear elastic region (b) the nonlinear region

To complete the nonlinear analysis, the cross section of the member will be broken up into several layers and the stresses and moments experienced by each layer will be calculated at its centroid. This allows the nonlinear material behaviour of the structure to be accounted for, as some regions of the cross section will yield before others. An additional benefit is that the cross section no longer needs to be homogeneous. Each layer can be allocated its own yield stress and elastic modulus, allowing a beam to me made up of different materials. However, a drawback in the development of this model is that two materials cannot be within the same layer, therefore, the number of layers must be selected such that the boundary of each material coincides with the edge of a layer. This can cause complexity in the analysis of a layered material.

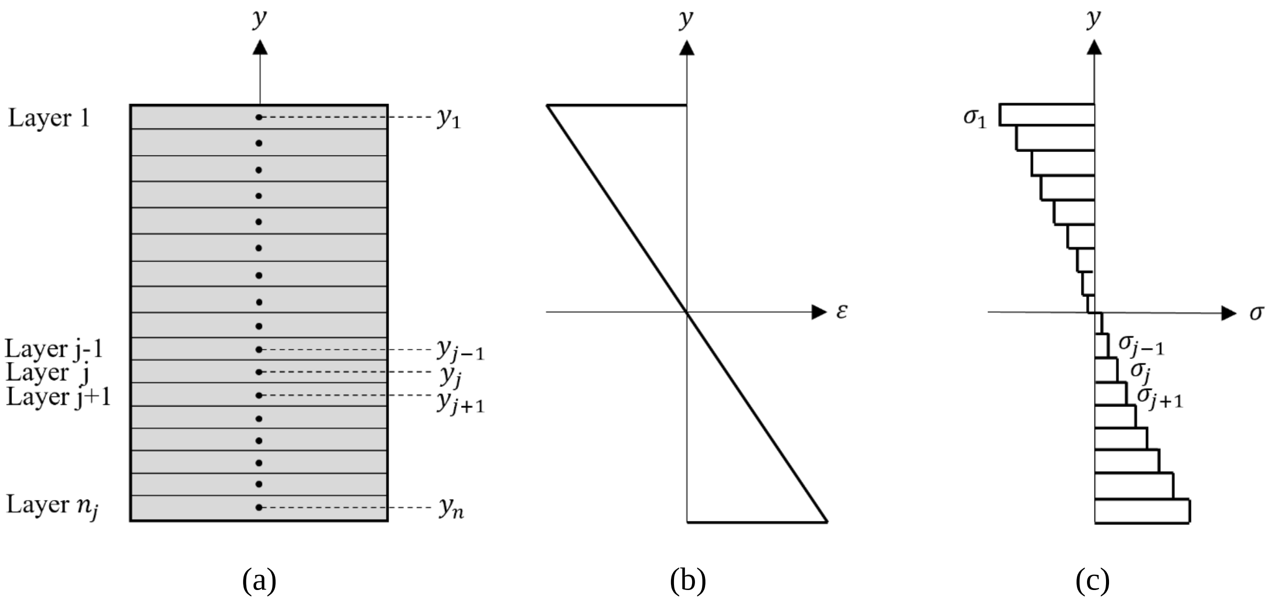


Figure 5.3: Cross sectional discretisation (a) Cross section (b) Strain diagram (c) Stress Diagram

Figure 5.3 presents the cross-sectional discretisation and shows that the stresses and strains in the top and bottom fibres of the cross section are the greatest of any point in the cross section. This means that these regions would reach yield at a lower applied loading than the points closer to the centre of the beam.

Figure 5.4 presents the difference in the stress diagram of a nonlinear cross section with a high number of layers to a cross section with a lower number of layers. It is evident that the cross section with a greater number of layers more closely represents the linear distribution of stress that would occur for a beam in the linear elastic range. As the size of the layer increases, the error increases because the stress is constant over a larger area.

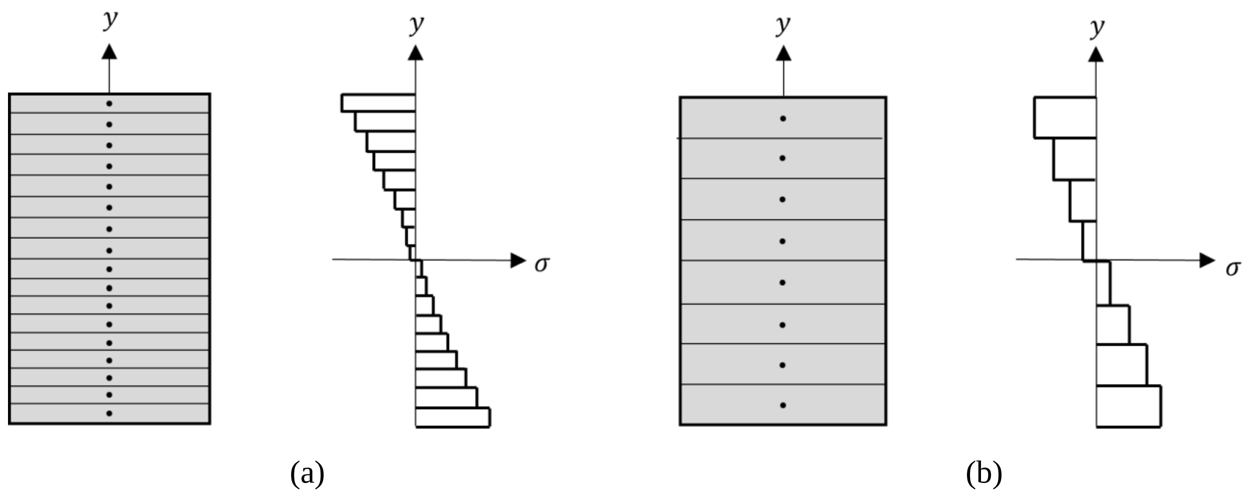


Figure 5.4: Cross section and stress diagram for a (a) high number of elements (b) low number of elements

When considering the top extreme layer, the use of a larger layer means that the centroid will be at a lower position than if a smaller layer was used. This would lead to a lower calculated strain value and therefore a lower calculated stress. This is unsafe for design as this means that the beam would experience yielding at a higher stress and ultimately have a higher assumed failure stress.

However, increasing the number of layers increases the computations required for the analysis as the stress must be evaluated at each layer for each cross section. Therefore, for efficiency a balance should be achieved between the level of accuracy and the computational power needed for the analysis.

The accuracy could also be increased by changing the assumption that the stress is constant for each layer. One possible method would be to assume a linear variation of the stress in each layer, however, this will not be explored in this thesis.

5.2: Newton-Raphson Method

The nonlinear analysis will be completed using the Newton-Raphson method, which is an iterative process that gives approximations closer to the solution after each iteration. A tolerance value is used to determine whether the solution has sufficiently converged and once this value is reached, the iterations conclude leaving the final approximation of the solution.

As for the linear analysis case, the behaviour of the member is described by the following relationship:

$$\boldsymbol{K}(\boldsymbol{D}) = \boldsymbol{Q} \tag{5.4}$$

where Q is the vector of applied loads and K(D) is a collection of the nonlinear functions that describe the internal actions.

At least two iterations must be completed to be able to determine whether convergence has been achieved. The solution at the next iteration is approximated using the first terms of the Taylor expansion of Equation 5.4 as follows:

$$\boldsymbol{K}(\boldsymbol{D}^{(i+1)}) \approx \boldsymbol{K}(\boldsymbol{D}^{(i)}) + \boldsymbol{K_t}(\boldsymbol{D}^{(i)})\Delta\boldsymbol{D}^{(i)} = \boldsymbol{Q} \tag{5.5}$$

where $\boldsymbol{K_t}(\boldsymbol{D}^{(i)})$ describes the tangent behaviour of the member, calculated using the displacement along the member and $\Delta\boldsymbol{D}^{(i)}$ is the vector of increments of the member displacements. Additionally, (i+1) denotes the solution of the next iteration.

It is useful to separate the terms involving $\Delta\boldsymbol{D}^{(i)}$, which gives the following equations:

$$\boldsymbol{K}_t(\boldsymbol{D}^{(i)})\Delta\boldsymbol{D}^{(i)} = \boldsymbol{Q_R} \tag{5.5}$$

$$\boldsymbol{Q}_R = \boldsymbol{Q} - \boldsymbol{K}(\boldsymbol{D}^{(i)}) \tag{5.6}$$

where $\boldsymbol{Q_R}$ is the vector of residual loads

The process for nonlinear analysis using the Newton Raphson method is presented visually in Figure 5.5. Firstly, an initial assumed displacement must be chosen to be used in the first iteration. For the problem in this thesis, it is assumed that there is initially zero displacement throughout the member. With this assumption, it is possible to determine the initial tangent stiffness $\boldsymbol{K_t}(\boldsymbol{D}^{(1)})$, which allows the displacement increment to be calculated using Equation 5.6. This is achieved by following the same procedure in Chapter 3, where the unknown and known displacements and loads are separated as shown in Equations 3.33 and 3.34.

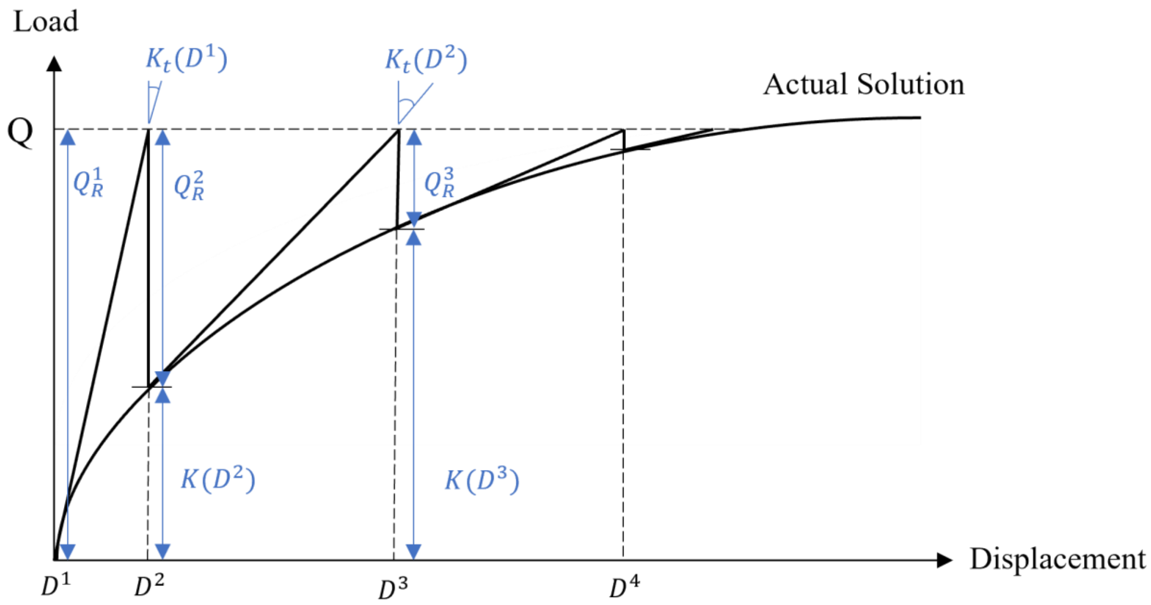


Figure 5.5: The Newton-Raphson method for nonlinear analysis

Once $\Delta\boldsymbol{D}^{(1)}$ is calculated, the initial guess of the displacement in the member can be updated with this displacement increment, giving new displacement values $\boldsymbol{D}^{(2)}$. To determine whether the solution has converged, the following norm will be used:

$$norm_{tol} = \frac{\left|\boldsymbol{Q}_{\boldsymbol{R}}^{(\boldsymbol{i}+\mathbf{1})}\right|}{|\boldsymbol{Q}|} \tag{5.7}$$

where the solution is assumed to have converged once this norm has reached a value lower than 10^{-5}. If the norm value is calculated to be greater than 10^{-5}, iterations are completed as required until the norm value is sufficiently reduced.

To create the nonlinear finite element model using the Newton-Raphson method, the displacements of the beam must be described using the nodal freedoms of the structure. **Q** and $\boldsymbol{K}(\boldsymbol{D}^{(i)})$ can be solved in a similar way to the linear analysis presented in Chapter 3. Equation 3.14 is extended to account for the material nonlinearity as presented in Appendix D, giving the following integrals:

$$\boldsymbol{k}_e(x, d_e) = \int_L \boldsymbol{B}^T(x)\boldsymbol{r}(x, d_e)\, dx \tag{5.8}$$

$$\boldsymbol{q}_e = \int_L N_e^T p\ dx \tag{5.9}$$

$$\boldsymbol{k}_{et}(x, d_e) = \int_L \boldsymbol{B}^T(x)\boldsymbol{r}_t(x, d_e)\, dx \tag{5.10}$$

where **B**, $\boldsymbol{N}_e$ and **p** can be found in Appendix E. Vectors $\boldsymbol{r}$ and $\boldsymbol{r}_t$ are given in Equations 5.17 and 5.18 respectively.

The Gauss-Legendre formulae are used to calculate the integrals for k_e, q_e and k_{et} across each finite element. For any interval [a, b], the integral of a function $f(x)$ is given by the following weighted summation:

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2}\sum_{k=1}^{n_G} w_k\, f\left(\frac{a+b}{2}+\frac{b-a}{2}\bar{x}_k\right) \tag{5.11}$$

If the polynomial has a degree of up to $(2n_G + 1)$ then the approximation of the integral is exact. Table 5.1 contains the weighting functions, w_k and function arguments x_k for different numbers of Gauss integration points n_G.

Table 5.1: Weighting functions and function arguments for different numbers of Gauss points [14]

n_G	$\bar{x}_k$	w_k
1	$x_1 = 0$	$w_1 = 2$
2	$x_1 = -0.577350269$ $x_2 = 0.577350269$	$w_1 = 1$ $w_2 = 1$
3	$x_1 = -0.774596669$ $x_2 = 0$ $x_3 = 0.774596669$	$w_1 = 0.5555556$ $w_2 = 0.8888888$ $w_3 = 0.5555556$
4	$x_1 = -0.861136312$ $x_2 = -0.339981044$ $x_3 = 0.339981044$ $x_4 = 0.861136312$	$w_1 = 0.3478548$ $w_2 = 0.6521452$ $w_3 = 0.6521452$ $w_4 = 0.3478548$
5	$x_1 = -0.906179846$ $x_2 = -0.538469310$ $x_3 = 0$ $x_4 = 0.538469310$ $x_5 = 0.906179846$	$w_1 = 0.2369269$ $w_2 = 0.04786287$ $w_3 = 0.5688888$ $w_4 = 0.4786287$ $w_5 = 0.2369269$
6	$x_1 = -0.932469514$ $x_2 = -0.661209386$ $x_3 = -0.238619186$ $x_4 = 0.238619186$ $x_5 = 0.661209386$ $x_6 = 0.932469514$	$w_1 = 0.1713245$ $w_2 = 0.3607616$ $w_3 = 0.4679139$ $w_4 = 0.4679139$ $w_5 = 0.3607616$ $w_6 = 0.1713245$

Using the weighted summation in Equation 5.11, Equations 5.8, 5.9 and 5.10 can be rewritten as:

$$\boldsymbol{k}_e(x, d_e) \approx \frac{L}{2}\sum_{k=1}^{n_G} w_k\, \boldsymbol{B}^T(x_k)\boldsymbol{r}(x_k, d_e) \tag{5.12}$$

$$\boldsymbol{q}_e \approx \frac{L}{2}\sum_{k=1}^{n_G} w_k\, \boldsymbol{N}_e^T(x_k)\boldsymbol{p}(\boldsymbol{x_k}) \tag{5.13}$$

$$\boldsymbol{k}_{et}(x, d_e) \approx \frac{L}{2}\sum_{k=1}^{n_G} w_k\, \boldsymbol{B}^T(x_k)\boldsymbol{r_t}(x_k, d_e) \tag{5.14}$$

where:

$$x_k = \frac{L}{2}(\bar{x}_k + 1) \tag{5.15}$$

and $\bar{x}_k$ and w_k are the function arguments and weighting functions for the Gauss integration points respectively and L is the length of the finite element.

r is the vector of internal actions and contains the internal axial force and moment for the cross section. Using the definitions of axial force and moment gives:

$$\boldsymbol{r}(x_k, d_e) = \begin{bmatrix} N(x_k, \boldsymbol{d}_e) \\ M(x_k, \boldsymbol{d}_e) \end{bmatrix} = \begin{bmatrix} \int_A \sigma(x_k, y, \boldsymbol{d}_e)\, dA \\ -\int_A y\sigma(x_k, y, \boldsymbol{d}_e)\, dA \end{bmatrix} \tag{5.16}$$

As previously discussed, the cross section will be discretised into n_j layers to allow the nonlinear material behaviour to be accounted for. The internal axial force and moment will be evaluated at the centroid of each layer, y_j.Therefore, the integrals for the internal actions can be approximated by the sum of the internal actions calculated for each layer. Therefore, Equation 5.16 can be rewritten as:

$$\boldsymbol{r}(x, d_e) = \begin{bmatrix} \sum_{j=1}^{n_j} \sigma(x, y_j, \boldsymbol{d}_e^{(i)})\, A_j \\ -\sum_{j=1}^{n_j} y_j\sigma(x, y_j, \boldsymbol{d}_e^{(i)})\, A_j \end{bmatrix} \tag{5.17}$$

where σ is evaluated according to Equations 5.1 to 5.3 and $\boldsymbol{d}_e^{(i)}$ denotes the element displacement vector for the current iteration.

To create the finite element model, the displacements at the degrees of freedom will be approximated by polynomials. Because the nonlinear analysis uses Euler Bernoulli beam theory, the equations for the horizontal and vertical displacements as well as the shape functions are the same as outlined in Equations 3.22 to 3.25.

$\boldsymbol{r}_t$ is given by evaluating the partial derivative of the internal actions with respect to $\boldsymbol{d}_e$, which is the vector containing the displacement values for each degree of freedom of the element under consideration. Therefore $\boldsymbol{r}_t$ can be expressed as:

$$\boldsymbol{r}_t = \begin{bmatrix} \frac{\partial N_k^{(i)}}{\partial d_{e1}} & \frac{\partial N_k^{(i)}}{\partial d_{e2}} & \frac{\partial N_k^{(i)}}{\partial d_{e3}} & \frac{\partial N_k^{(i)}}{\partial d_{e4}} & \frac{\partial N_k^{(i)}}{\partial d_{e5}} & \frac{\partial N_k^{(i)}}{\partial d_{e6}} & \frac{\partial N_k^{(i)}}{\partial d_{e7}} \\ \frac{\partial M_k^{(i)}}{\partial d_{e1}} & \frac{\partial M_k^{(i)}}{\partial d_{e2}} & \frac{\partial M_k^{(i)}}{\partial d_{e3}} & \frac{\partial M_k^{(i)}}{\partial d_{e4}} & \frac{\partial M_k^{(i)}}{\partial d_{e5}} & \frac{\partial M_k^{(i)}}{\partial d_{e6}} & \frac{\partial M_k^{(i)}}{\partial d_{e7}} \end{bmatrix} \tag{5.18}$$

where $N_k^{(i)}$and $M_k^{(i)}$ denote the internal axial force and moment calculated using the function argument x_k for the current iteration and d_{e1}through to d_{e7} are the elemental displacements for the local freedoms 1 through 7. The equations for $\frac{\partial N_k^{(i)}}{\partial d_e}$ and $\frac{\partial M_k^{(i)}}{\partial d_e}$ are given in Appendix F.

Chapter 6: Validation of Models

Chapters 3, 4 and 5 have outlined the process of creating finite elements for the linear models of EBBT and TBT as well as a nonlinear model for EBBT. These finite elements have been implemented into Matlab so that the calculations can be completed efficiently. The models created can output a variety of information defining the beam behaviour such as the curvature, horizontal displacement and vertical displacement when material, geometrical and load parameters are input into the model. To ensure the Matlab models have no bugs and output appropriate results, it is important to compare them against the output of other analysis techniques. The deflection output of the linear models will be validated against Strand 7 whereas the nonlinear model will be compared against the output of the analytical nonlinear solution.

6.1: Comparison with Strand 7

The first case to be validated is the deflection for a beam with a high length to depth ratio. 100 elements were used to ensure convergence was achieved for both the Euler-Bernoulli and Timoshenko beam models. The beam modelled is presented in Figure 6.1. The beam has an elastic modulus of 20000 MPa and a shear modulus of 8000 MPa.

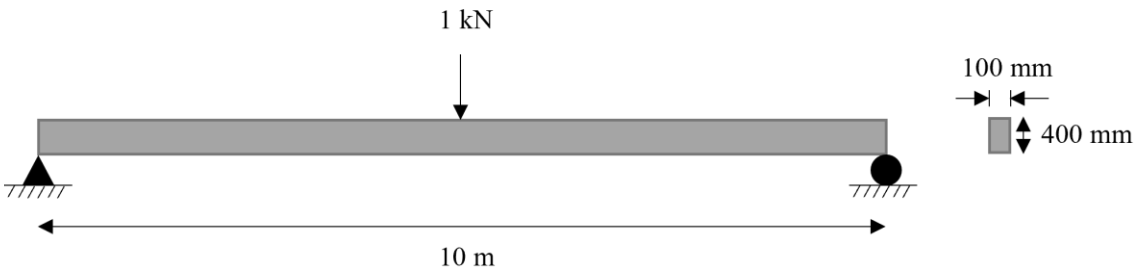


Figure 6.1: Elevation and cross section of beam used for validation of linear models

Figure 6.2 presents a comparison between the displacement output of Strand7 and the Matlab models for EBBT and TBT. Only the vertical displacement within the central half of the beam is shown. This is done so that the differences between the beam models can be seen clearly. For the entire length of the beam, the output of the EBBT beam model is coincident with the Strand7 output. The deflection output obtained from the TBT beam model was marginally higher than for the other two models, however, due to the additional shear deformations accounted for in TBT this is the expected result. Overall, it is evident that the models have been created properly for this scenario.

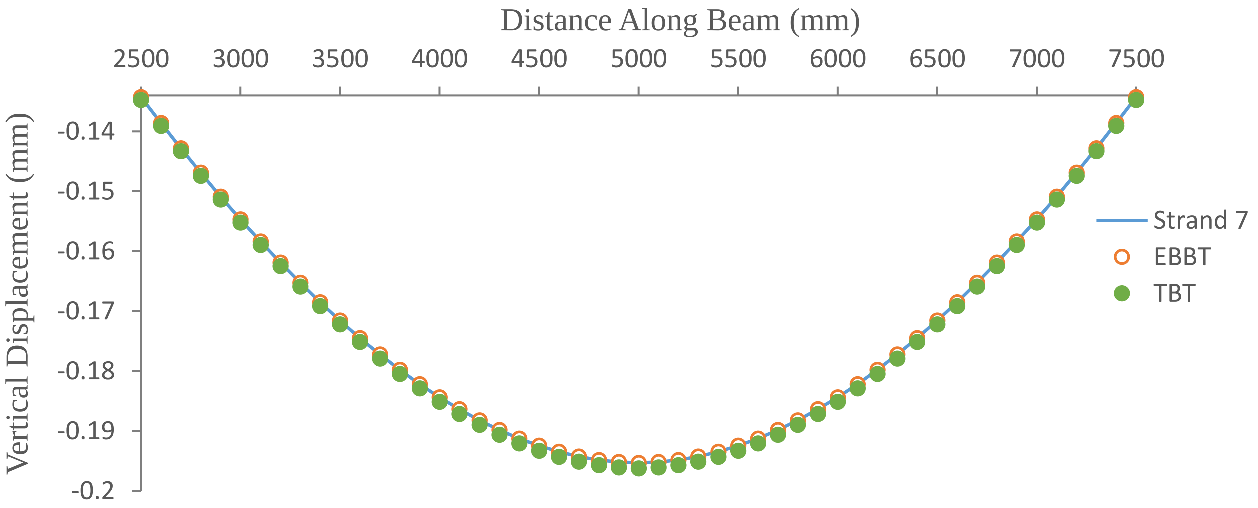


Figure 6.2: Comparison between deflection output of the Matlab finite element models and Strand7

6.2: Comparison with Analytical Solution

The nonlinear analysis Matlab model will be validated against the analytical solution which is also known as the closed form solution (CFS). The closed form solution is presented in Appendix G. The load is applied such that the yield stress is exceeded to ensure that the model performs appropriately. The loading and geometric parameters used are presented in Figure 6.3. The material has an elastic modulus of 200 GPa and a yield strength of 300 MPa. For the Matlab model, 100 elements and 100 layers are used to ensure that convergence of the solution has occurred.

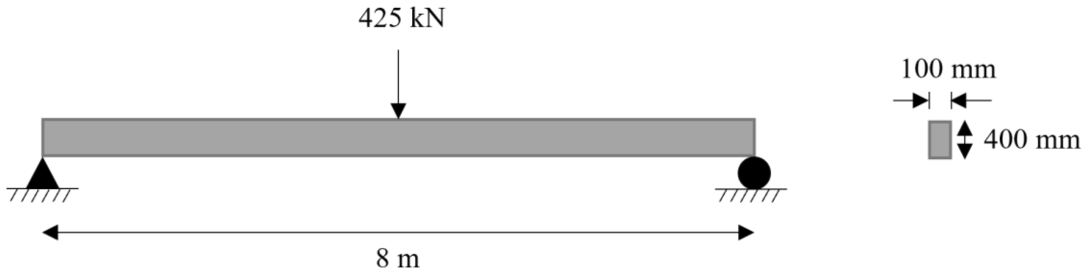


Figure 6.3: Elevation and cross section of beam used for validation of nonlinear model

Figure 6.3 shows that the Matlab nonlinear model almost exactly aligns with the CFS at all points along the beam. Therefore, it is accepted that the nonlinear Matlab model has been created properly.

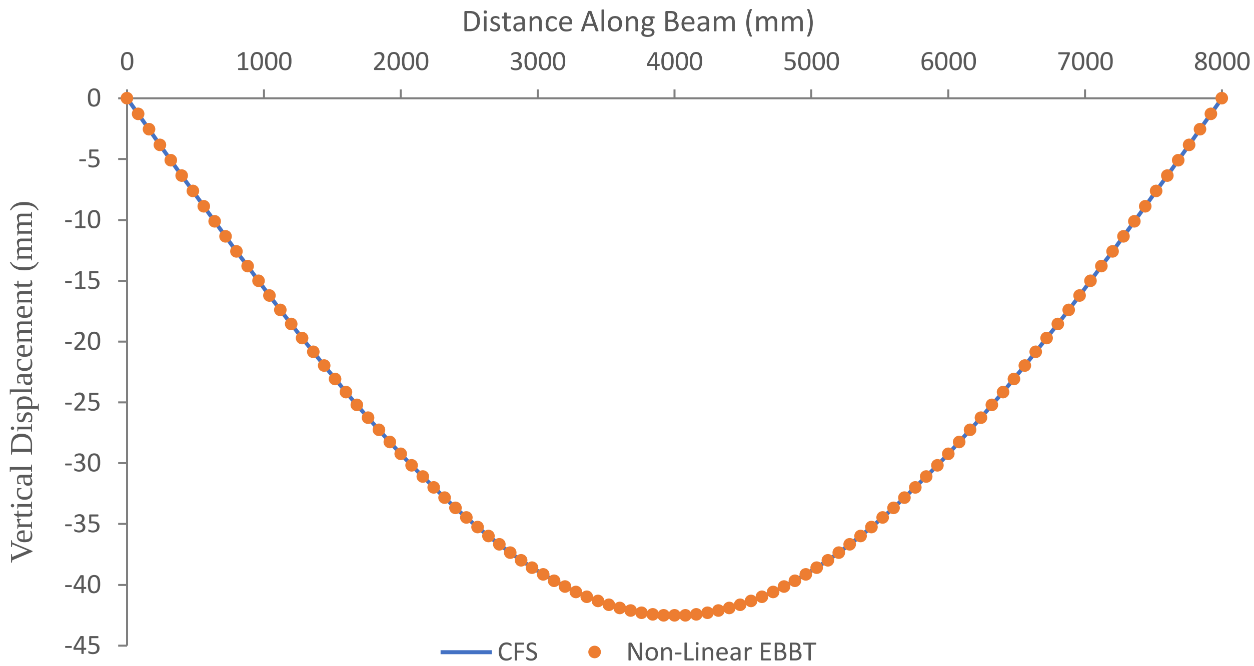


Figure 6.4: Comparison between the output of the closed form solution (CFS) and the nonlinear Matlab model

Chapter 7: Results and Discussion

This thesis aims to explore the differences in the behaviour of beams modelled using different finite element models. In this chapter, the three finite element models that were developed in Chapters 3 to 5 will be compared to determine the suitability of each model to different scenarios. Additionally, factors such as the number of elements and number of layers will be investigated to determine their impact on the accuracy of the solution achieved.

7.1 Number of elements

One significant decision that must be made before completing finite element analysis is the level of discretisation that will be completed. In many scenarios, increasing the number of elements that the beam is broken down into will lead to a more accurate result. However, increasing the number of elements increases the number of calculations that must be completed, raising the computational demand of the analysis.

Thus, the selection of the number of elements used during finite element analysis is a process that requires a balance between computational cost and accuracy. It is useful to determine the number of elements that are needed for the solution to converge, thus allowing for an efficient number of elements to be chosen. This is because once the solution has converged, an increase in the number of elements will not result in a significant change in output of the model. The Matlab code for each beam model will be run for a varied number of elements to investigate the differences in the convergence rates between each beam model.

To investigate the linear analysis models, the loading scenario and beam properties are the same as for the beam shown in Figure 6.1. Figure 7.1 compares the percentage error of the deflection for a different number of elements for both the EBBT and TBT models. The deflection at the midpoint of the supported beam was compared against a beam with 100 elements as the solution is assumed to have converged at this level of discretisation. The rate of convergence of the Timoshenko beam model was found to be slower than for the Euler-Bernoulli beam model, requiring 30 elements to achieve a change of less than 0.01% between element jumps. On the other hand, the model for EBBT converged to the exact solution with just two elements.

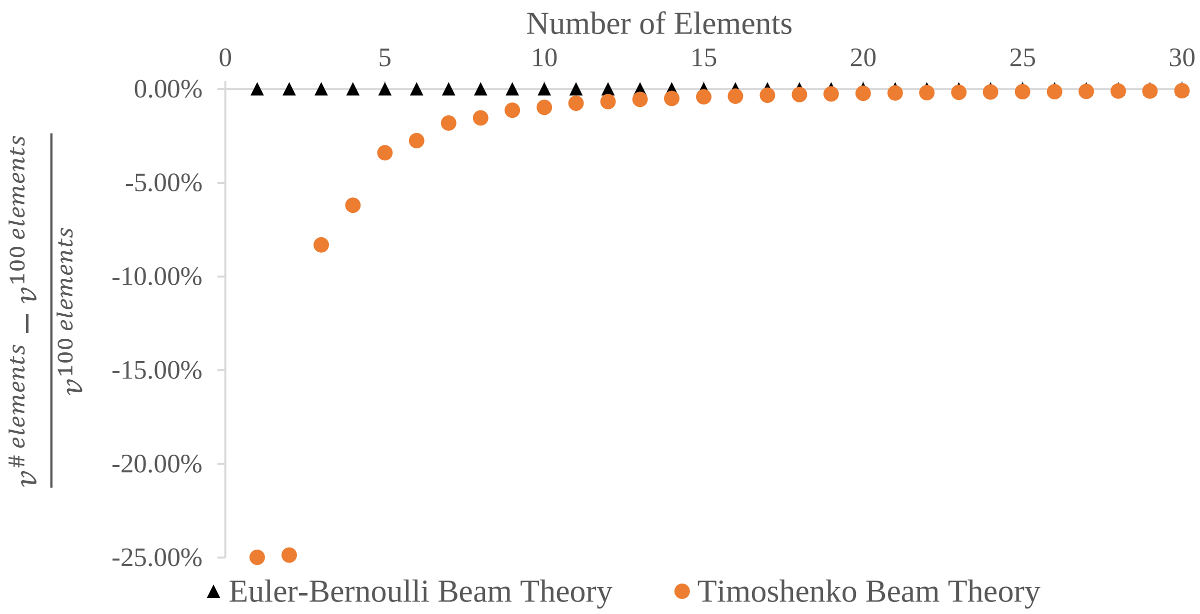


Figure 7.1: Comparison of the mid-span deflection for EBBT and TBT for different levels of mesh refinement

This highlights the importance of checking the convergence of the solution as while running the analysis with two elements would output an accurate beam deflection using the Euler-Bernoulli beam model, the results of the Timoshenko model would be underestimated by 25%. Additionally, it can be surmised that the use of a Euler-Bernoulli beam model is more computationally efficient as it can be run using a coarser discretisation than the Timoshenko beam model.

It was also investigated how the convergence rate differs depending on the structural response variable being output by the model. The beam curvature is an important result a designer would be interested in obtaining from the analysis. Figure 7.1 compares the convergence rate for mid span curvature and deflection using the Euler-Bernoulli beam model.

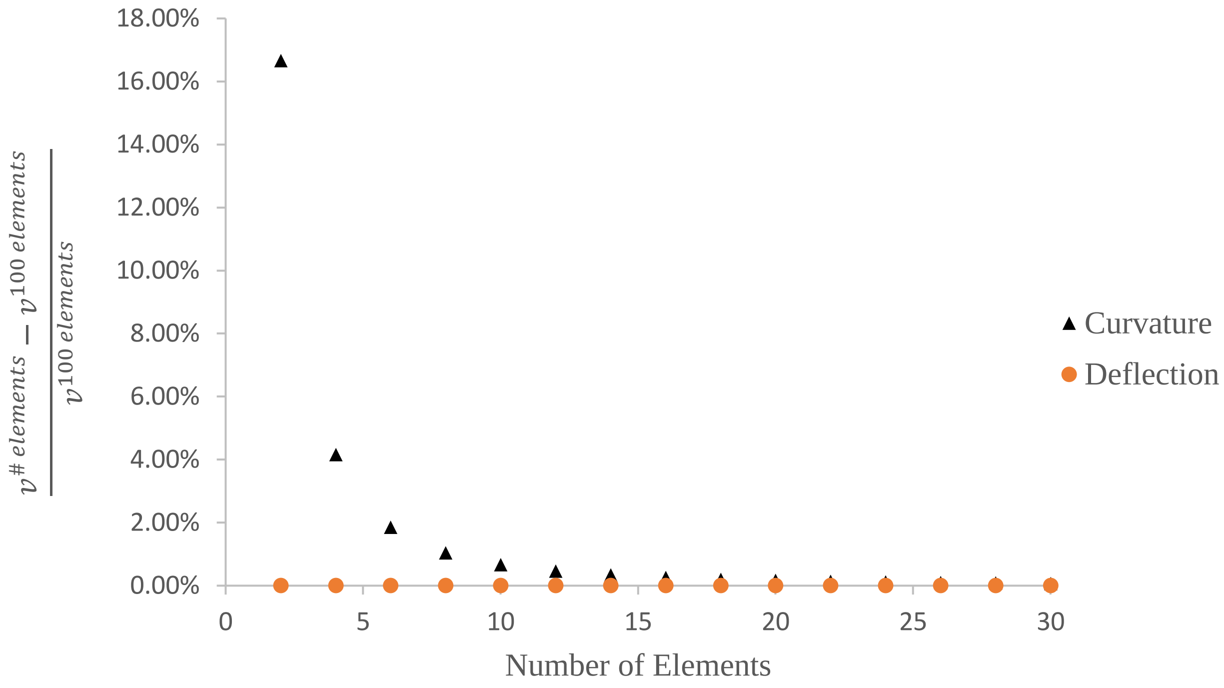


Figure 7.2: Comparison of the convergence of the mid-span deflection and curvature for linear EBBT

While the use of two elements would be enough to obtain an accurate deflection value, there would still be a 16.7% error in the curvature. This is because the curvature is the second derivative and therefore requires a finer discretisation to achieve convergence. Therefore, it is important to check the convergence rate for all the variables of interest to make sure that an accurate result is obtained.

A similar analysis must be conducted for the nonlinear analysis. Figure 7.3 presents the percentage error of the mid span deflection and curvature of a beam against the number of elements used in the nonlinear analysis. The mid span values were compared against those of the closed form solution presented in Appendix G. The results indicate that a greater number of elements are required for the nonlinear analysis when compared to the EBBT linear analysis model presented in Figure 7.2. Similarly to case of the linear analysis, the mid span curvature required more elements to achieve convergence than the mid span deflection, needing 94 and 20 elements to achieve a change less than 0.01% between element jumps respectively.

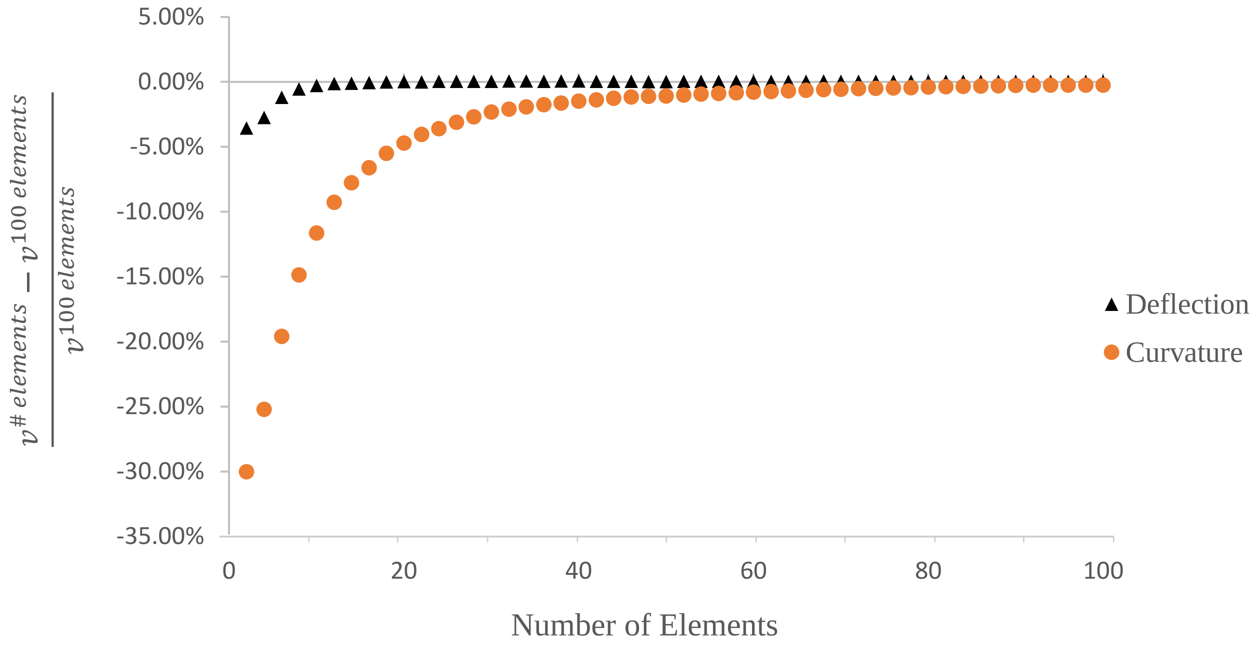


Figure 7.3: Comparison of the convergence of the mid-span deflection and curvature for nonlinear EBBT

7.2 Length to Depth Ratio

The length to depth (L/D) ratio gives an indication of the degree to which shear deformations contribute to the overall displacement of the beam. The lower the L/D ratio is, the more shear dominated the beam displacement is. The Euler-Bernoulli and Timoshenko beam models were tested to explore how the displacements obtained from the models differed from each other for a variety of L/D ratios. It could then be determined whether the shear deformations are significant enough to warrant their consideration and if they are significant, which L/D ratios they remain dominant for.

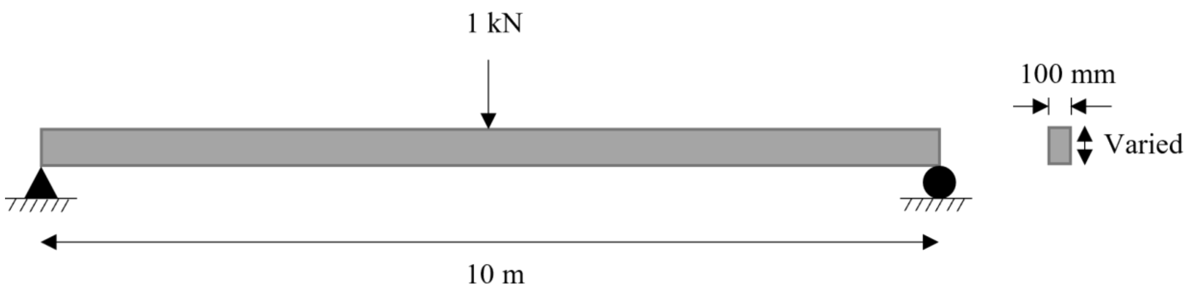


Figure 7.4: Elevation and cross section of the beam used for the L/D analysis

The beam under consideration is presented in Figure 7.4. A comparison between the mid span deflection of the Euler-Bernoulli beam model and the Timoshenko beam model for different length to depth ratios is presented in figure 7.5. As expected, there is a significant percentage difference between the mid span deflections of the two beam theories for low length to depth ratios. This is because shear deformations are significant for deep beams, causing the Timoshenko beam model to output a greater value of mid span deflection. On the other hand, as the Euler-Bernoulli beam model does not have the capability to account for shear deformations it underrepresented the deflections for deep beams. At an L/D ratio of 1, the Euler-Bernoulli beam model underestimates the deflection by 75%, which a significant level of error. This shows the importance of understanding the underlying assumptions of the model being used for analysis, as Euler Bernoulli beam theory is unsafe for use in the design of deep beams.

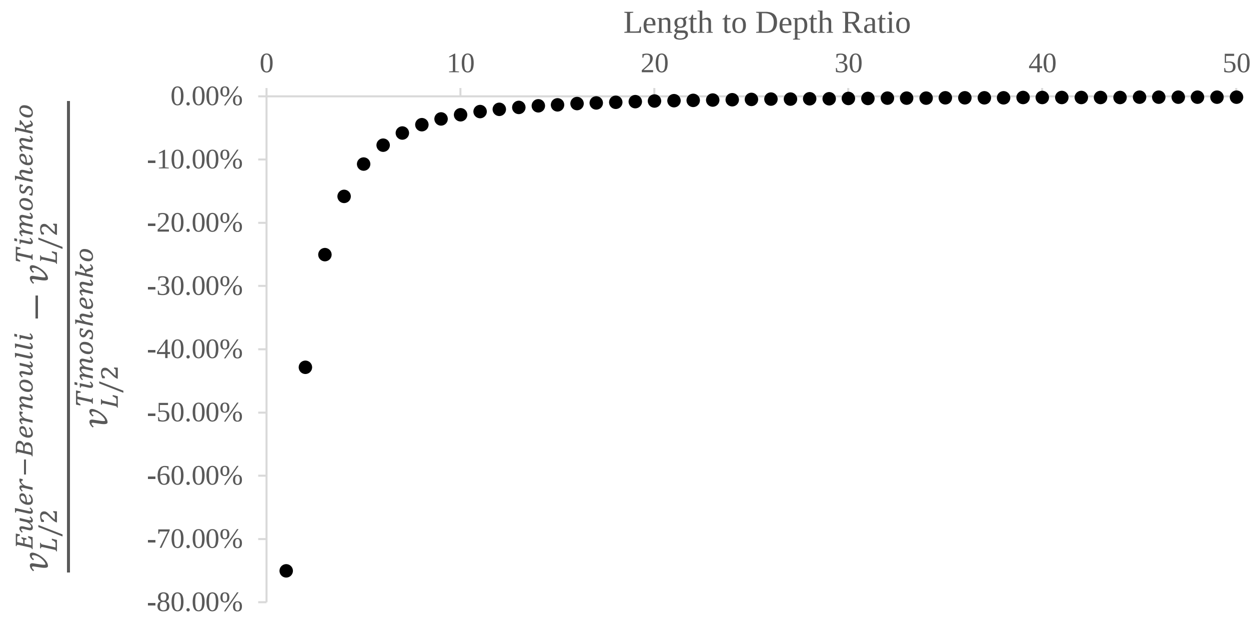


Figure 7.5: Comparison of the mid-span deflection for EBBT and TBT for different L/D ratios

The difference between the two theories reduces as the L/D ratio is increased. This is because flexural deformations begin to dominate and the shear deformations become negligible, allowing the output of the Timoshenko beam model to converge to that of the Euler-Bernoulli beam model. The rate of change in the difference between the models is significant up until a L/D ratio of about 20, where the Timoshenko beam model begins to converge to the solution of the Euler-Bernoulli beam model. Once the L/D ratio hits 50, the difference between the deflection output drops below 0.1%.

7.3 Number of layers

For nonlinear analysis, both the cross section and the member must be discretised for analysis. The influence of the number of elements chosen for the member analysis has already been investigated in Chapter 7.1. The number of layers that are used in the cross-sectional analysis is significant because the stress and moment within each layer are constant and calculated at the centroid of the layer. Therefore, a smaller layer would allow the distribution of stress throughout the cross section to be more realistically modelled. However, increasing the number of layers significantly increases the computational requirements of the analysis. This is because the axial force and moment values must be calculated for each layer of the cross section for every element of the beam for every iteration completed for the analysis. As a result, even slight increases to the number of layers will significantly increase the number of calculations required to obtain the solution.

The number of layers required for convergence was compared against the closed form solution presented in Appendix G for the mid span deflection and curvature. The percentage difference between the mid span deflection in the finite element model and the closed formed solution for different numbers of layers is presented in Figure 7.6. The solution is shown to converge to the analytical solution once there are around 40 layers.

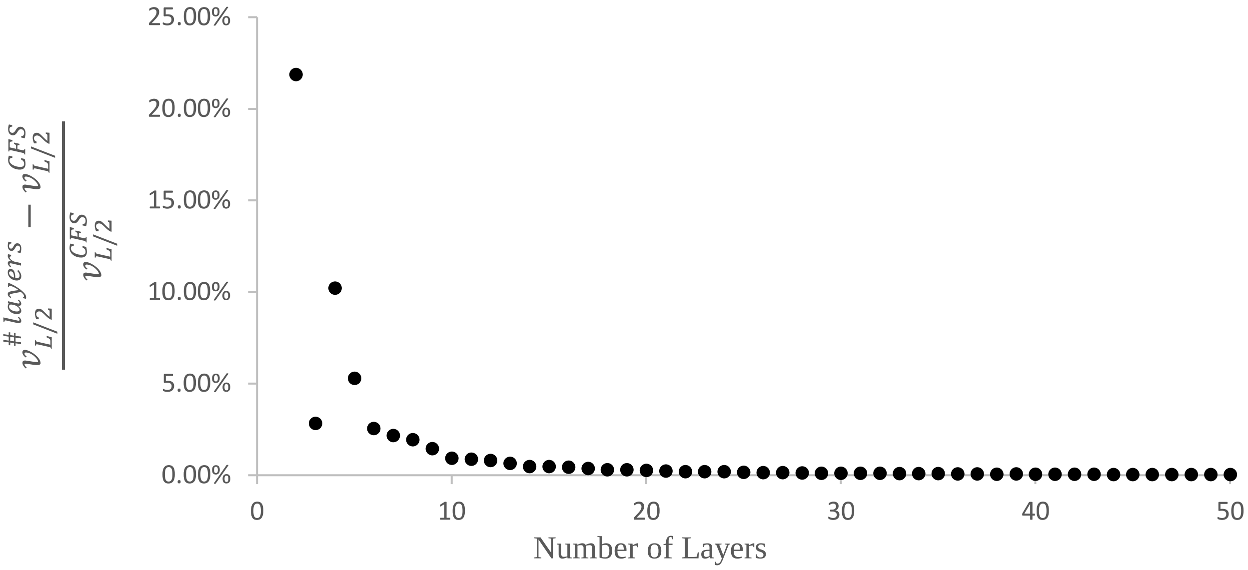


Figure 7.6: Convergence of the mid-span deflection for nonlinear EBBT

The percentage difference between the mid span curvature in the finite element model and the closed formed solution for different numbers of layers is presented in Figure 7.7. For fewer than 30 layers, the change in curvature is highly unstable, fluctuating between values higher and lower than the analytical solution. By 40 layers, the oscillations conclude, and the solution converges.

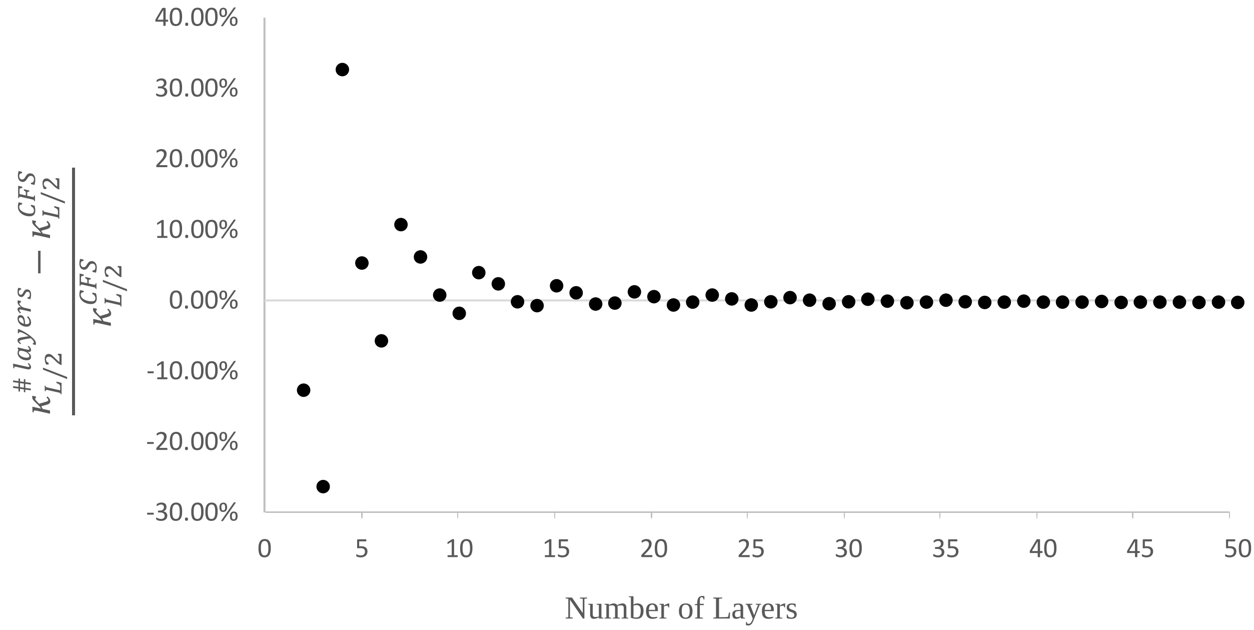


Figure 7.7: Convergence of the mid-span curvature for nonlinear EBBT

For both the curvature and deflection, 40 layers provides an adequate level of accuracy for the nonlinear finite element analysis model. Therefore, for the analysis moving forward, 40 layers and 100 elements will be used.

7.5 Linear analysis vs Nonlinear Analysis

The beam used to compare the linear and nonlinear EBBT models is shown in Figure 7.8. The beam has a yield strength of 300 MPa and an elastic modulus of 200 GPa.

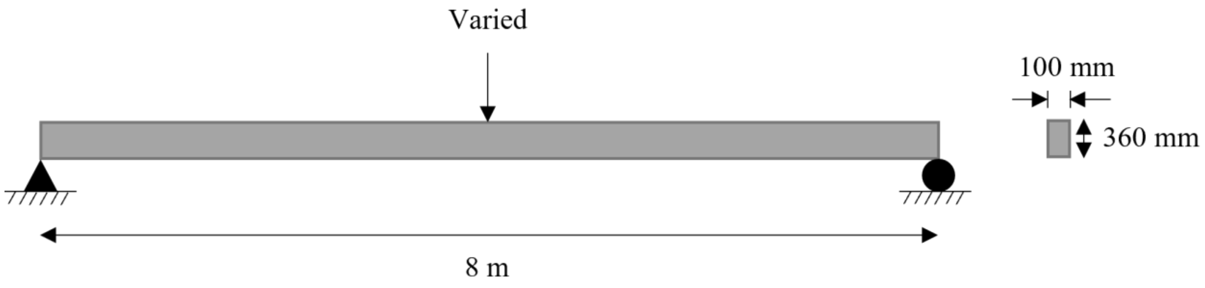


Figure 7.8: Elevation and cross section of the beam used for the comparison of the linear and nonlinear Euler-Bernoulli beam models

The benefit of nonlinear analysis is that it can explore the behaviour of a partially yielded cross section. This is clearly shown in Figure 7.9, which compares the mid span displacement of the linear and nonlinear Euler-Bernoulli beam models. The output of both models are equal until the applied load reaches 350 kN at which point the stress in the extreme layers of the cross section begin to yield. At this point, using the linear EBBT model is no longer valid. The significance of this is seen as the load continues to increase and more of the cross section has yielded.

The linear model assumes that the stress is still given by Hooke's law and therefore underestimates the deflection of the beam. This can be seen as once the load increased beyond 325 kN the difference between the mid span deflections increases at an increasing rate. For the nonlinear beam model, once yielding begins the beam will experience significant deformation without a large increase to the applied loading as presented in the constitutive model in Figure 5.1. As the outer layers begin to yield, the stress is redistributed to the layers closer to the centre and the full capacity of the cross section is realised. The model failed to run once the load passed 475 kN as this is when the entire cross section had yielded. This provides a capacity 150 kN greater than if only a linear analysis was run.

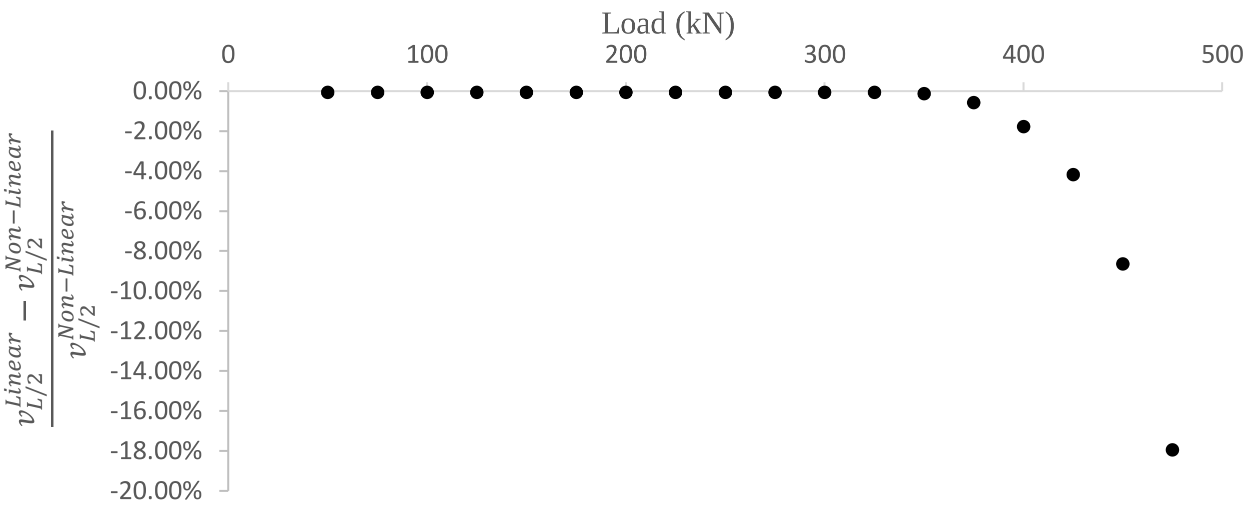


Figure 7.9: Percentage difference between the mid-span deflection of the linear and nonlinear Euler-Bernoulli beam models as the load is varied

When completing a linear analysis, it is important to check that the stress in the critical cross section has not exceeded the yield strength of the material. Otherwise, the vertical deflection of the beam could be significantly underestimated. When the applied loading was 475 kN, the linear Euler-Bernoulli beam model underestimated the displacement by 18%. This could lead to an unsafe design that does not meet strength and serviceability requirements.

Chapter 8: Conclusion

From the analysis in Chapter 7, it is clear that each of the Matlab finite element models developed performs differently depending on factors such as the beam geometry and loading conditions. It has been highlighted that the use of a model that is not properly suited to the situation can result in significant underestimations of displacements.

For low ratios of length to depth, the Timoshenko beam model was shown to present the beam deformations more accurately. Euler-Bernoulli underrepresented the vertical displacements by up to 75%, which is a considerable error. However, once the L/D ratio reached 50 the difference between the output of the two models became negligible. Thus, it was shown that for low L/D ratios, the Euler-Bernoulli beam model is not an appropriate model choice. However, once the beam was sufficiently long, Euler Bernoulli provided a good approximation of the beam displacement while being a simpler model than the Timoshenko beam model. This was the expected result and corresponds to the theory outlined in Chapter 2 because it was demonstrated that deep beams experience significant shear deformations that cannot be accounted for using the Euler-Bernoulli beam model.

Convergence analysis was also completed for all three beam models. It was found that the Euler-Bernoulli beam model could be run using a coarser discretisation than the Timoshenko beam model. Thus, it would be more computationally efficient to use the Euler-Bernoulli beam model if the beams being analysed are sufficiently long despite both models providing similar results.

It was also shown that different variables will have different convergence rates. For the linear Euler-Bernoulli beam theory, it was found that the curvature required a finer discretisation than the deflection. Thus, careful consideration should be made by the designer to ensure that the convergence is checked for each of the variables they are investigating. It was also found that in the linear elastic region, nonlinear analysis required a finer discretisation of the member to achieve convergence. Thus, despite being shown to provide the same results as the linear model, it is more efficient to use the linear Euler-Bernoulli beam model within the linear elastic range.

Unlike the linear models, nonlinear analysis involves discretisation of the cross section into several layers. The convergence rate was found to be similar for the curvature and deflection, however, the curvature results fluctuated significantly for a low number of layers. This shows that a designer must ensure that both the cross section and the member are sufficiently discretised to achieve an accurate output from the model.

The difference between the output of the linear and nonlinear Euler-Bernoulli beam models was also investigated to observe the significance of the nonlinear behaviour. It was found that once the beam began to yield, the linear Euler-Bernoulli model significant underestimated the deflection, with up to an 18% error. The ability for nonlinear analysis to give an insight into the full capacity of a beam was also realised. Once the beam yielded, the stress was redistributed to the central portion of the cross section that would usually only be subjected to small loads in the elastic range. It was found that the beam in Figure 7.8 had a 475 kN capacity with the nonlinear analysis, whereas the linear analysis was only valid to a maximum applied load of 325 kN.

Overall, the results show the importance that a designer understands the underlying assumptions of the model they are using and the types of beams they are appropriate for. It is also important that when using finite element models, the level of discretisation required to attain convergence is checked for each of the beam variables of interest. This is to ensure that an accurate and valid result is obtained.

Further research:

The beam models created in this thesis were relatively simple and can be enhanced to analyse more complicated cross sections or increase the accuracy of the results obtained. A key part of the development of the finite element model was using polynomials to approximate the displacements throughout the beam. This thesis used elements with seven degrees of freedom as this was the simplest possible element that could meet consistency requirements and prevent shear locking from occurring. However, more accurate approximations of the displacement could be used by using higher order polynomials, and therefore, a higher degree of freedom element.

Additionally, the linear models assumed that the beam was made from an isotropic and homogeneous material, which means that they cannot be used to explore the behaviour of materials that have different material properties in different directions. Extending the work of this thesis to account for anisotropic materials such as timber would enhance the applicability of these models.

As previously discussed, for the nonlinear analysis the internal actions are evaluated at the centroid of each layer and are considered constant throughout the cross section. Improvements can be made to the accuracy of the analysis by instead assuming that these actions vary linearly within the cross section. Another way to improve the accuracy of the nonlinear model is to use a more complicated constitutive model to depict the nonlinear behaviour of the beam.

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Appendix A: Derivation of Stiffness Matrix and Loading Vector

From Equation 3.14:

$$\int_L \boldsymbol{D}\hat{\boldsymbol{\varepsilon}} \cdot \boldsymbol{A}\hat{\boldsymbol{e}}\, dx \;=\; \int_L \boldsymbol{p} \cdot \hat{\boldsymbol{e}}\, dx \tag{A1}$$

where

$$\boldsymbol{D} \;=\; \begin{bmatrix} EA & -EB \\ -EB & EI \end{bmatrix} \tag{A2}$$

$$\boldsymbol{\varepsilon} \;=\; \begin{bmatrix} \varepsilon_r \\ \kappa \end{bmatrix} = \begin{bmatrix} u' \\ v'' \end{bmatrix} = \begin{bmatrix} \partial & 0 \\ 0 & \partial^2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \mathbf{Ae} \tag{A3}$$

where A is a differential operator, the symbol ∂ represents differentiation with respect to x and:

$$\boldsymbol{e} \;=\; \begin{bmatrix} u \\ v \end{bmatrix} \tag{A4}$$

The member loads n and w are collected in p:

$$\boldsymbol{p} \;=\; \begin{bmatrix} n \\ w \end{bmatrix} \tag{A5}$$

From Equations 3.22 and 3.24, u and v have been approximated as:

$$u \;=\; N_{u1}u_L + N_{u2}u_M + N_{u3}u_R \tag{A6}$$

$$v \;=\; N_{v1}v_L + N_{v2}\theta_L + N_{v3}v_R + N_{v4}\theta_R \tag{A7}$$

which can be rewritten in compact form as:

$$\boldsymbol{e} \;=\begin{bmatrix} u \\ v \end{bmatrix} \;=\; \begin{bmatrix} N_{u1} & 0 & 0 & N_{u2} & N_{u3} & 0 & 0 \\ 0 & N_{v1} & N_{v2} & 0 & 0 & N_{v3} & N_{v4} \end{bmatrix} \begin{bmatrix} u_L \\ v_L \\ \theta_L \\ u_M \\ u_R \\ v_R \\ \theta_R \end{bmatrix} \;=\; \boldsymbol{N}_e \boldsymbol{d}_e \tag{A8}$$

Therefore, Equation A3 can be rewritten as:

$$\boldsymbol{\varepsilon} = \mathbf{B}\boldsymbol{d}_e \tag{A9}$$

where

$$\mathbf{B} \;=\; \mathbf{A}\boldsymbol{N}_e \tag{A10}$$

$$\mathbf{B} \;=\; \begin{bmatrix} -\frac{3}{L}+\frac{4x}{L^2} & 0 & 0 & \frac{4}{L}-\frac{8x}{L^2} & -\frac{1}{L}+\frac{4x}{L^2} & 0 & 0 \\ 0 & \frac{12x}{L^3}-\frac{6}{L^2} & \frac{6x}{L^2}-\frac{4}{L} & 0 & 0 & \frac{6}{L^2}-\frac{12x}{L^3} & \frac{6x}{L^2}-\frac{2}{L} \end{bmatrix} \tag{A11}$$

Substituting Equations A8 and A9 into Equation A1 gives:

$$\int_L \boldsymbol{D}\mathbf{B}\boldsymbol{d}_e \cdot \boldsymbol{B}\widehat{\boldsymbol{d}}_e \, dx \; = \; \int_L \boldsymbol{p} \cdot \boldsymbol{N}_e \widehat{\boldsymbol{d}}_e \, dx \tag{A12}$$

The virtual displacements $\widehat{\boldsymbol{d}}_e$ are then isolated by using the fact that $\boldsymbol{Aa} \cdot \boldsymbol{Bb} \; = \; \boldsymbol{B^T Aa} \cdot \boldsymbol{b}$ as follows:

$$\int_L \boldsymbol{B^T D}\mathbf{B}\boldsymbol{d}_e dx \cdot \widehat{\boldsymbol{d}}_e \;\; = \; \int_L \boldsymbol{N}_e^T \boldsymbol{p} dx \cdot \widehat{\boldsymbol{d}}_e \tag{A13}$$

From which the stiffness relationship of the finite element can be obtained:

$$k_e d_e = q_e \tag{A14}$$

where:

$$k_e = \int_L \boldsymbol{B^T D}\mathbf{B} \, dx \tag{A15}$$

$$q_e = \int_L \boldsymbol{N}_e^T \boldsymbol{p} \, dx \tag{A16}$$

Integrating along the member length gives:

$$k_e \;\; = \begin{bmatrix} \frac{7EA}{3L} & \frac{-4\mathrm{EB}}{L^2} & \frac{-3\mathrm{EB}}{L} & \frac{-8EA}{3L} & \frac{EA}{3L} & \frac{4\mathrm{EB}}{L^2} & \frac{-\mathrm{EB}}{L} \\ \frac{-4\mathrm{EB}}{L^2} & \frac{12EI}{L^3} & \frac{6EI}{L^2} & \frac{8\mathrm{EB}}{L^2} & \frac{-4\mathrm{EB}}{L^2} & \frac{-12EI}{L^3} & \frac{6EI}{L^2} \\ \frac{-3\mathrm{EB}}{L} & \frac{6EI}{L^2} & \frac{4EI}{L} & \frac{4\mathrm{EB}}{L} & \frac{-\mathrm{EB}}{L} & \frac{-6EI}{L^2} & \frac{2EI}{L} \\ \frac{-8EA}{3L} & \frac{8\mathrm{EB}}{L^2} & \frac{4\mathrm{EB}}{L} & \frac{16EA}{3L} & \frac{-8EA}{3L} & \frac{-8\mathrm{EB}}{L^2} & \frac{4\mathrm{EB}}{L} \\ \frac{EA}{3L} & \frac{-4\mathrm{EB}}{L^2} & \frac{-\mathrm{EB}}{L} & \frac{-8EA}{3L} & \frac{7EA}{3L} & \frac{4\mathrm{EB}}{L^2} & \frac{-3\mathrm{EB}}{L} \\ \frac{4\mathrm{EB}}{L^2} & \frac{-12EI}{L^3} & \frac{-6EI}{L^2} & \frac{-8\mathrm{EB}}{L^2} & \frac{4\mathrm{EB}}{L^2} & \frac{12EI}{L^3} & \frac{-6EI}{L^2} \\ \frac{-\mathrm{EB}}{L} & \frac{6EI}{L^2} & \frac{2EI}{L} & \frac{4\mathrm{EB}}{L} & \frac{-3\mathrm{EB}}{L} & \frac{-6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \tag{A17}$$

$$q_e = \begin{bmatrix} \frac{L}{6}n & \frac{L}{2}w & \frac{L^2}{12}w & \frac{2L}{3}n & \frac{L}{6}n & \frac{L}{2}w & \frac{-L^2}{12}w \end{bmatrix}^T \tag{A18}$$

Appendix B: Consistency Requirements for EBBT

For the Euler-Bernoulli beam model, the strain expression is:

$$\varepsilon_x = u' - yv'' \qquad \text{(B1)}$$

Therefore, to achieve consistency, $u'\ and\ v''$ must be of the same order. There are three horizontal degrees of freedom, therefore u is approximated by a parabolic function:

$$u = a_o + a_1 x + a_2 x^2 \qquad \text{(B2)}$$

There are four degrees of freedom associated with deflection, therefore v is approximated by a cubic polynomial:

$$v = b_0 + b_1 x + b_2 x^2 + b_3 x^3 \qquad \text{(B1)}$$

Therefore, $u'\ and\ v''$ are given by:

$$u' = a_1 + 2a_2 x \qquad \text{(B2)}$$

$$v' = b_1 + 2b_2 x + 3b_3 x^2 \qquad \text{(B5)}$$

$$v'' = 2b_2 + 6b_3 x \qquad \text{(B6)}$$

$u'\ and\ v''$ are both first order functions, therefore the consistency requirement is satisfied.

Appendix C: TBT matrices and vectors

$$r = \begin{bmatrix} N \\ M \\ S \end{bmatrix} = D\varepsilon \tag{C1}$$

$$D = \begin{bmatrix} EA & -EB & 0 \\ -EB & EI & 0 \\ 0 & 0 & AG \end{bmatrix} \tag{C2}$$

$$\varepsilon = \begin{bmatrix} \varepsilon_r \\ \kappa \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} u' \\ \theta' \\ v' - \theta \end{bmatrix} = \begin{bmatrix} \partial & 0 & 0 \\ 0 & 0 & \partial \\ 0 & \partial & -1 \end{bmatrix} \begin{bmatrix} u \\ v \\ \theta \end{bmatrix} = Ae \tag{C3}$$

$$A = \begin{bmatrix} \partial & 0 & 0 \\ 0 & 0 & \partial \\ 0 & \partial & -1 \end{bmatrix} \tag{C4}$$

$$e = \begin{bmatrix} u \\ v \\ \theta \end{bmatrix} \tag{C5}$$

$$\mathrm{p} = \begin{bmatrix} n \\ w \\ 0 \end{bmatrix} \tag{C6}$$

Appendix D: Weak Form of Nonlinear

$$\int_L \boldsymbol{D}\hat{\boldsymbol{\varepsilon}} \cdot \boldsymbol{A}\hat{\boldsymbol{e}}\, dx = \int_L \boldsymbol{p} \cdot \hat{\boldsymbol{e}}\ dx \tag{D1}$$

$$\boldsymbol{r} = \boldsymbol{D}\boldsymbol{\varepsilon} \tag{D2}$$

where:

$$\boldsymbol{r} = \begin{bmatrix} N \\ M \end{bmatrix} \qquad \boldsymbol{D} = \begin{bmatrix} EA & -EB \\ -EB & EI \end{bmatrix} \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_r \\ \kappa \end{bmatrix} \tag{D3a-c}$$

The generalised displacements e can be approximated by polynomial functions as follows:

$$\boldsymbol{e} \approx \boldsymbol{N}_e \boldsymbol{d}_e \tag{D4}$$

where N_e are the shape functions given in equations 3.10 and 3.11 and d_e is the vector of nodal displacements

Additionally, the strain field can be expressed in terms of the nodal displacements as follows:

$$\boldsymbol{\varepsilon} = \boldsymbol{A}\boldsymbol{N}_e \boldsymbol{d}_e \tag{D5}$$

$$\boldsymbol{B} = \boldsymbol{A}\boldsymbol{N}_e \tag{D6}$$

Substituting equation AF.2, AF.4 and AF.6 into equation AF.1 gives:

$$\int_L \boldsymbol{r} \cdot \boldsymbol{B}\hat{\boldsymbol{d}}_e\, dx = \int_L \boldsymbol{p} \cdot \boldsymbol{N}_e \hat{\boldsymbol{d}}_e\ dx \tag{D7}$$

Using the fact that $Aa \cdot Bb = B^T Aa \cdot b$, Equation D7 can be rewritten as:

$$\int_L \boldsymbol{B}^T \boldsymbol{r} \cdot \hat{\boldsymbol{d}}_e\, dx = \int_L \boldsymbol{N}_e^T \boldsymbol{p} \cdot \hat{\boldsymbol{d}}_e\ dx \tag{D8}$$

The stiffness relationship for each finite element is given by:

$$\boldsymbol{k}_e(\boldsymbol{d}_e) = \boldsymbol{q}_e \tag{D9}$$

Therefore:

$$\boldsymbol{k}_e(\boldsymbol{d}_e) = \int_L \boldsymbol{B}^T \boldsymbol{r}\, dx \tag{D10}$$

$$\boldsymbol{q}_e = \int_L \boldsymbol{N}_e^T\ \boldsymbol{p}\ dx \tag{D11}$$

Appendix E: Nonlinear EBBT Matrices

$$N_e = \begin{bmatrix} N_{u1} & 0 & 0 & N_{u2} & N_{u3} & 0 & 0 \\ 0 & N_{v1} & N_{v2} & 0 & 0 & N_{v3} & N_{v4} \end{bmatrix} \tag{E1}$$

$$\boldsymbol{B} = \begin{bmatrix} N'_{u1}(x_k) & 0 & 0 & N'_{u2}(x_k) & N'_{u3}(x_k) & 0 & 0 \\ 0 & N''_{v1}(x_k) & N''_{v2}(x_k) & 0 & 0 & N''_{v3}(x_k) & N''_{v4}(x_k) \end{bmatrix} \tag{E2}$$

$$\boldsymbol{p} = \begin{bmatrix} n \\ w \end{bmatrix} \tag{E3}$$

Appendix F: Components of $\boldsymbol{r_t}$

$$\frac{\partial N_k^{(i)}}{\partial d_{e1}} \approx \sum_{j=1}^{n_j} A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left(-\ \frac{3}{L} + \frac{4x_k}{L^2}\right) \tag{F1}$$

$$\frac{\partial N_k^{(i)}}{\partial d_{e2}} \approx \sum_{j=1}^{n_j} A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left[-\, y_j\left(\frac{12x_k}{L^3} - \frac{6}{L^2}\right)\right] \tag{F2}$$

$$\frac{\partial N_k^{(i)}}{\partial d_{e3}} \approx \sum_{j=1}^{n_j} A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left[-\, y_j\left(\frac{6x_k}{L^2} - \frac{4}{L}\right)\right] \tag{F3}$$

$$\frac{\partial N_k^{(i)}}{\partial d_{e4}} \approx \sum_{j=1}^{n_j} A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left(\frac{4}{L} - \frac{8x_k}{L^2}\right) \tag{F4}$$

$$\frac{\partial N_k^{(i)}}{\partial d_{e5}} \approx \sum_{j=1}^{n_j} A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left(-\ \frac{1}{L} + \frac{4x_k}{L^2}\right) \tag{F5}$$

$$\frac{\partial N_k^{(i)}}{\partial d_{e6}} \approx \sum_{j=1}^{n_j} A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left[-\, y_j\left(\frac{6}{L^2} - \frac{12x_k}{L^3}\right)\right] \tag{F6}$$

$$\frac{\partial N_k^{(i)}}{\partial d_{e7}} \approx \sum_{j=1}^{n_j} A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left[-\, y_j\left(\frac{6x_k}{L^2} - \frac{2}{L}\right)\right] \tag{F7}$$

$$\frac{\partial M_k^{(i)}}{\partial d_{e1}} \approx -\sum_{j=1}^{n_j} y_j A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left(-\ \frac{3}{L} + \frac{4x_k}{L^2}\right) \tag{F8}$$

$$\frac{\partial M_k^{(i)}}{\partial d_{e2}} \approx \sum_{j=1}^{n_j} y_j^2 A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left(\frac{12x_k}{L^3} - \frac{6}{L^2}\right) \tag{F9}$$

$$\frac{\partial M_k^{(i)}}{\partial d_{e3}} \approx \sum_{j=1}^{n_j} y_j^2 A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left(\frac{6x_k}{L^2} - \frac{4}{L}\right) \tag{F10}$$

$$\frac{\partial M_k^{(i)}}{\partial d_{e4}} \approx -\sum_{j=1}^{n_j} y_j A_j \ \frac{\partial \sigma(x_k,\ y_j, d_e^{(i)}}{\partial \varepsilon}\left(\frac{4}{L} - \frac{8x_k}{L^2}\right) \tag{F11}$$

$$\frac{\partial M_k^{(i)}}{\partial d_{e5}} \approx -\sum_{j=1}^{n_j} y_j A_j \frac{\partial \sigma(x_k, y_j, d_e^{(i)}}{\partial \varepsilon}\left(-\frac{1}{L} + \frac{4x_k}{L^2}\right) \tag{F12}$$

$$\frac{\partial M_k^{(i)}}{\partial d_{e6}} \approx \sum_{j=1}^{n_j} y_j^2 A_j \frac{\partial \sigma(x_k, y_j, d_e^{(i)}}{\partial \varepsilon}\left(\frac{6}{L^2} - \frac{12x_k}{L^3}\right) \tag{F13}$$

$$\frac{\partial M_k^{(i)}}{\partial d_{e7}} \approx \sum_{j=1}^{n_j} y_j^2 A_j \frac{\partial \sigma(x_k, y_j, d_e^{(i)}}{\partial \varepsilon}\left(\frac{6x_k}{L^2} - \frac{2}{L}\right) \tag{F14}$$

Appendix G: Closed Form Solution

For a simply supported beam subjected to a central point load:

In the linear elastic region of the beam [14]:

$$\theta_{Linear} = \frac{Px^2}{4EI} + \frac{3M_y^2\beta_1\beta_3 + \beta_2\beta_3\left(4M_p^2 - 4M_pM_y\right) + 2\beta_1\beta_2M_p\left(PL - 4M_p\right)}{-3P\beta_1\beta_3EI} \quad \text{(G1)}$$

$$v_{Linear} = \frac{Px^3}{12EI} + \left(\frac{3M_y^2\beta_1\beta_3 + \beta_2\beta_3\left(4M_p^2 - 4M_pM_y\right) + 2\beta_1\beta_2M_p\left(PL - 4M_p\right)}{-3P\beta_1\beta_3EI}\right)x \quad \text{(G2)}$$

In the nonlinear region of the beam [14]:

$$\theta_{Nonlinear} = -\frac{2}{3}\frac{\sqrt{6M_p}\kappa_y}{P}(2M_p - Px)^{\frac{1}{2}} + \frac{2\sqrt{3}\kappa_y\beta_3}{3P} \quad \text{(G3)}$$

$$v_{Nonlinear} = \frac{4}{9}\frac{\sqrt{6M_p}\kappa_y}{P^2}(2M_p - Px)^{\frac{3}{2}} + \left(\frac{2\sqrt{3}\kappa_y\beta_3}{3P}\right)x - \frac{4[M_y^2(3M_y\beta_1 - 2M_p\beta_2) + 2M_p^2\beta_2\left(2M_p - M_y\right)}{9\beta_1P^2EI} \quad \text{(G4)}$$

where

$$\beta_1 = \sqrt{(M_p - M_y)M_p} \quad \text{(G5)}$$

$$\beta_2 = \sqrt{3}\kappa_yEI \quad \text{(G6)}$$

$$\beta_3 = \sqrt{(4M_p - PL)M_p} \quad \text{(G7)}$$

$$\kappa_y = \frac{2f_y}{Ed} \quad \text{(G8)}$$